SIEMENS

SINUMERIK 840D/840Di/810D

Short Guide Programming **12.2000 Edition**

ISO Dialects for SINUMERIK

Manufacturer/Service Documentation

SIEMENS

SINUMERIK 840D/840Di/810D

ISO Dialects for SINUMERIK

Short Guide Programming

Valid for

Control	*Software version*
SINUMERIK 840D	6
SINUMERIK 840DE (export version)	6
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Preface

Reader notes

The SINUMERIK documentation is divided into three parts:

- General Documentation
- User Documentation
- Manufacturer/Service Documentation

For more detailed information on SINUMERIK 840D publications and other publications covering all SINUMERIK controls, please contact your local SIEMENS office.

Subject of the manual

The information contained in this manual gives details of how to read in part programs from external CNC systems and how to execute them.

For whose use is the manual?

- Programmers
- Servicing and operating personnel

Safety notes

The following notes are given to ensure your safety and to prevent any risk of damage to the product described or units in the machines connected.



Warning

When electrical equipment is in operation, certain parts of such equipment are bound to reach hazardous voltage levels.

Work on the device/system by **unqualified** personnel or failure to comply with warnings can result in severe injury or material damage. Only suitably **qualified personnel** trained in the setting up, installation, commissioning or operation of the product should carry out work on the device/system.

Safety-related references

This manual serves as a guideline for your personal safety as well as for avoiding property damage. Your attention is drawn to safety-related references by a warning triangle detailing the danger level as follows:

Proper use

Please take note of the following:

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Warning

The unit must be used only for such purposes as specified in the catalog and in the technical description and only in connection with non-Siemens units and components recommended or approved by Siemens.

Trouble-free and safe operation of the product can only be guaranteed by proper transport and storage, installation and set-up as well as by careful operation and maintenance.



Danger

indicates an imminently hazardous situation which, if not avoided, **will** result in death or serious injury.



Warning

indicates a potentially hazardous situation which, if not avoided, **could** result in death or serious injury.



Caution

used with the safety alert symbol indicates a potentially hazardous situation which, if not avoided, **may** result in minor or moderate injury.

Caution

used without safety alert symbol indicates a potentially hazardous situation which, if not avoided, **may** result in property damage.

Notice

used without the safety alert symbol indicates a potential situation which, if not avoided, **may** result in an undesireable result or state.

Note

Important information on the product, handling of the product or the respective part of the documentation to which your attention is specifically drawn.

Qualified staff

Only qualified staff are allowed to start up and operate a unit. Qualified staff in the sense of the safety-related references referred to in this manual are members of personnel authorized to commission, earth and mark up units, systems and electric circuits according to the standards applying to safety technology.

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Programming 1

1.1 Activation of functions

Machine data 18800 $MN_EXTERN_LANGUAGE is used to activate the external language. The language type, ISO Dialect–M or T is selected with machine data 10880 $MN_EXTERN_CNC_SYSTEM.
The external language can be activated separately for each channel.
For example, channel 1 can operate in ISO mode but channel 2 is active in Siemens mode.

Switchover

The following two G commands from Group 47 are used to switch between Siemens mode and ISO Dialect mode:

- G290 Siemens NC programming language active
- G291 ISO Dialect NC programming language active

The active tool, tool offset and zero offset remain active here (see Subsection 1.2.2 and Section 1.4).

Siemens mode

The following conditions apply when Siemens mode is active:

- Siemens G commands are interpreted on the control by default.
- It is not possible to extend the Siemens programming system with ISO Dialect functions because some of the G functions have different meanings.
- Downloadable MD files can be used to switch the control to ISO Dialect mode. In this case, the user sees the ISO Dialect mode by default.

ISO Dialect mode

The following conditions apply when ISO Dialect mode is active:

- Only ISO Dialect G codes can be programmed, not Siemens G codes.
- It is not possible to use a mixture of ISO Dialect code and Siemens code in the same NC block.
- It is not possible to switch between ISO Dialect-M and ISO Dialect-T via G command
- If further Siemens functions are to be used, it is necessary to switch to Siemens mode first (exception: program branching and subprogram calls, see Subsection 1.3.2)

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Power ON/Reset

Table 1-1 shows the possible combinations of machine data $MN_EXTERN_CNC_SYSTEM and $MC_GCODE_RESET_VALUE. This specifies the Power ON/Reset response.

Table 1-1 Activation of functions

After Power ON/Reset...	$MC_GCODE_RESET_VALUES[46] =	$MN_EXTERN_CNC_SYSTEM =
Siemens mode active, switchover to ISO Dialect-M possible	**1** G290 Siemens mode	**1** ISO Dialect-M
Siemens mode active, switchover to ISO Dialect-T possible	**1** G290 Siemens mode	**2** ISO Dialect-T
ISO Dialect–M active, switchover to Siemens mode possible	**2** G291 ISO Dialect mode	**1** ISO Dialect-M
ISO Dialect–T active, switchover to Siemens mode possible	**2** G291 ISO Dialect mode	**2** ISO Dialect-T

Modal G commands

Modal G commands which have the same function in both systems (Siemens and ISO Dialect) are treated as follows.
When these G codes are programmed in one language, the equivalent G code in the other language is determined and activated. The following G codes are affected:

Data storage

ISO programs can be read into the MMC 103 and output in punchtape format. ISO programs which have been read in are stored in the NC data management system as main programs in the default path: _N_WKS_DIR/_N_SHOPMILL_WPD.

You can change the entry by editing the file DINO.INI in the USER directory. You will find further information in the publication

References: /IAM/, IM3: MMC Installation and Start-Up Guide. Section 3.1.

1.1.1 Switchover from ISO mode to Siemens mode

G290/291

G commands 290/291 can be used to change mode from the part program. On switchover, the display of current G codes also changes

G65/66

Non-modal and modal macro:
The programmed subprogram is called. Switchover to Siemens mode only takes place when the PROC instruction is used in the first line of the subprogram.
If a program of this type is terminated with M17 or RET, when the subprogram returns, the mode is switched back to ISO mode.

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Siemens subprogram call in ISO mode

Modal and non-modal subprogram calls, e.g.

```
    N100 CALL “SHAFT”
or
    N100 MCALL SHAFT
or
    N100 SHAFT
```

Modal and non-modal subprogram calls with parameter passing

```
    N100 MCALL SHAFT(”ABC”, 33.5)  or
    N100 SHAFT(“ABC”, 33.5)
```

Subprogram calls with path name

```
    N100 CALL “/_N_SPF_DIR/SHAFT    or
    N100 MCALL /_N_SPF_DIR/SHAFT    or
    N100 PCALL /_N_SPF_DIR/SHAFT
```

Siemens mode is selected implicitly on subprogram calls, and the system is switched back to ISO Dialect mode at the end of the subprogram.

Modal, non-modal cycles

If a modal or non-modal cycle is programmed in ISO mode, a shell cycle will be called.
This call results in switchover to Siemens mode.

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1.2 G commands

The G codes of ISO Dialect-T refer to G code system B (see also LEERER MERKER).

The active G codes in ISO mode can be read using system variable $P_EXTGG[...]. The numbers alongside the G code specify the respective value in $P_EXTGG[...]. Machine data 20154 EXTERN_GCODE_RESET_VALUES[n]: 0, ..., 30 is used to specify the G codes that are effective on start-up when the NC channel is not operating in Siemens mode.

Table 1-2 The default setting is indicated by 1)

ISO Dialect-T		ISO Dialect-M		Description
Group 1				
G00 1)	1	G00 1)	1	Rapid traverse
G01	2	G01	2	Linear motion
G02	3	G02	3	Circle/helix, clockwise
		G02.2	6	Evolvent, clockwise
G03	4	G03	4	Circle/helix, counterclockwise
		G03.2	7	Evolvent, counterclockwise
G33	5	G33	5	Thread cutting with constant lead
G34	9			Thread cutting with variable lead
G77	6			Longitudinal turning cycle
G78	7			Thread cutting cycle
G79	8			Face turning cycle
Group 2				
		G17 1)	1	XY plane
		G18	2	ZX plane
		G19	3	YZ plane
G96	1			Constant cutting rate ON
G97 1)	2			Constant cutting rate OFF
Group 3				
G90 1)	1	G90 1)	1	Absolute programming
G91	2	G91	2	Incremental programming
Group 4				
		G22	1	Working area limitation, protection zone 3 ON
		G23 1)	2	Working area limitation, protection zone 3 OFF
Group 5				
		G93	3	Inverse-time feedrate (rpm)
G94	1	G94 1)	1	Feed in [mm/min, inch/min]

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Table 1-2 The default setting is indicated by 1), continued

ISO Dialect-T		ISO Dialect-M		Description
G95 1)	2	G95	2	Feed in [mm/rev, inch/rev]
Group 6				
G20 1)	1	G20 1) (G70)	1	Input system inch
G21	2	G21 (G71)	2	Input system metric
Group 7				
G40 1)	1	G40 1)	1	Deselect cutter radius compensation
G41	2	G41	2	Compensation to left of contour
G42	3	G42	3	Compensation to right of contour
Group 8				
		G43	1	Tool length compensation positive ON
		G44	2	Tool length compensation negative ON
		G49 1)	3	Tool length compensation OFF
Group 9				
G22	1			Working area limitation, protection zone 3 ON
G23	2			Working area limitation, protection zone 3 OFF
		G73	1	Deep hole drilling cycle with chipbreaking
		G74	2	Counterclockwise tapping cycle
		G76	3	Fine drilling cycle
		G80 1)	4	Cycle off
		G81	5	Counterbore drilling cycle
		G82	6	Countersink drilling cycle
		G83	7	Deep hole drilling cycle with chip removal
		G84	8	Clockwise tapping cycle
		G85	9	Drilling cycle
		G86	10	Drilling cycle, retraction with G00
		G87	11	Reverse countersinking
		G89	13	Drilling cycle, retraction with machining feed
Group 10				
G80 1)	1			Drilling cycle off
G83	2			Face deep hole drilling
G84	3			Face tapping
G86	4			Face drilling
G87	5			Side deep hole drilling
G88	6			Side tapping
G89	7			Side drilling

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Table 1-2 The default setting is indicated by 1), continued

ISO Dialect-T		ISO Dialect-M		Description
		G98 1)	1	Return to starting point for fixed cycles
		G99	2	Return to point R for fixed cycles
Group 11				
G98 1)	1			Return to starting point for drilling cycles
G99	2			Return to point R for drilling cycles
		G50 1)	1	Scaling off
		G51	2	Scaling on
Group 12				
G66	1	G66	1	Modal macro call
G67 1)	2	G67 1)	2	Delete modal macro call
Group 13				
		G96	1	Constant cutting rate ON
		G97 1)	2	Constant cutting rate OFF
Group 14				
G54	1	G54 1)	1	Select zero offset
G55	2	G55	2	Select zero offset
G56	3	G56	3	Select zero offset
G57	4	G57	4	Select zero offset
G58	5	G58	5	Select zero offset
G59	6	G59	6	Select zero offset
		G54 P{1...48}1		Extended zero offsets
		G54 P0	1	“External ZO EXOFS”
Group 15				
		G61	1	Exact stop modal
		G63	2	Tapping mode
		G64 1)	3	Continuous-path mode
Group 16				
G17	1	G17 1)	1	XY plane
G18 1)	2	G18	2	ZX plane
G19	3	G19	3	YZ plane
		G68	1	Rotation ON 2D 3D
		G69	2	Rotation OFF
Group 17				
		G15 1)	1	Polar coordinates OFF
		G16	2	Polar coordinates ON

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Table 1-2 The default setting is indicated by 1), continued

ISO Dialect-T		ISO Dialect-M		Description
Group 18 (non-modal)				
G04	1	G04	1	Dwell
G05	20	G05	18	High-speed cycle cutting
G07.1	18	G07.1	16	Cylindrical interpolation
		G08	12	Feedforward control ON/OFF
		G09	2	Exact stop
G10	2	G10	3	Write zero offset/tool offset
G10.6	19			Rapid lift ON/OFF
		G11	4	Terminate parameter input
G27	16	G27	13	Referencing check (available soon)
G28	3	G28	5	Approach 1st reference point
G30	4	G30	6	Approach 2nd, 3rd, 4th reference point
G30.1	21	G30.1	19	Floating reference position
G31	5	G31	7	Measurement with touch-trigger probe
G52	6	G52	8	Additive zero offset
G53	17	G53	9	Approach position in machine coordinate system
G65	7	G65	10	Call macro
G70	8			Finishing cycle
G71	9			Stock removal cycle longitudinal axis
G72	10			Stock removal cycle transverse axis
		G72.1	14	Contour repetition, rotation
		G72.2	15	Contour repetition, linear
G73	11			Repeat contour
G74	12			Deep hole drilling and recessing in longitudinal axis (Z)
G75	13			Deep hole drilling and recessing in facing axis (X)
G76	14			Multiple thread cutting cycle
G92	15	G92	11	Preset actual value memory, spindle speed limitation
Group 20				
G50.2	1			Synchronous spindle OFF
G51.2	2			Synchronous spindle ON
Group 21				
G13.1	1			TRANSMIT OFF
G12.1	2			TRANSMIT ON
Group 22				
		G50.1	1	Mirroring on programmed axis OFF
		G51.1	2	Mirroring on programmed axis ON

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Table 1-2 The default setting is indicated by 1), continued

ISO Dialect-T		ISO Dialect-M		Description
Group 25				
		G131	1	Polar coordinates, interpolation
		G121	2	Polar coordinates, interpolation
Group 31				
G290 1)	1	G290 1)	1	Select Siemens mode
G291	2	G291	2	Select ISO Dialect mode

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Table 1-3 G commands are functionally identical in Siemens mode and in ISO Dialect mode

G commands in Siemens mode	Corresponding G commands in ISO Dialect-T	Corresponding G commands in ISO Dialect-M
Group 1: G00, G01, G02, G03, G33	Group 1: G00, G01, G02, G03, G33	Group 1: G00, G01, G02, G03, G33
Group 6: G17, G18, G19	Group 16: G17, G18, G19	Group 2: G17, G18, G19
Group 7: G40, G41, G42	Group 7: G40, G41, G42	Group 7: G40, G41, G42
Group 8: G54 to G554		Group 14: G54 to G59, G54 P1 to P48
Group 10: G60, G64		Group 15: G60, G64
Group 13: G700, G710	Group 6: G20, G21	Group 6: G20, G21
Group 14: G90, G91	Group 3: G90, G91	Group 3: G90, G91
Group 15: G94 G95 G96 G961 G97 G971	Group 5: G94 Group 2: G97 Group 5: G95 Group 2: G97 Group 5: G95 Group 2: G96 Group 5: G94 Group 2: G96 Group 5: G95 Group 2: G97 Group 5: G94 Group 2: G97	Group 5: G94 Group 13: G97 Group 5: G95 Group 13: G97 Group 5: G95 Group 13: G96 Group 5: G94 Group 13: G96 Group 5: G95 Group 13: G97 Group 5: G94 Group 13: G97

Note

If individual G codes of the groups in Table 1-3 cannot be mapped, the default setting in machine data
$MC_EXTERN_GCODE_RESET_VALUES and/or
$MC_GCODE_RESET_VALUES
is activated.

Example: `ISO mode`

```
N5  G00 X100. Y100.
N10 G90          ;Activate G90 in ISO mode Group 3
                 ;In Siemens mode Group 14

N15 G290         ;Switch over to Siemens, G90 is active
N20 G91          ;Activate G91 in ISO mode Group 3
                 ;In Siemens mode Group 14

N25  G291        ;Switch over to ISO mode
N30  G291        ;G91 is active
```

1.2.1 G code display

In the G code display, the G codes for the currently active language are displayed. G290/G291 also causes the G code display to switch over.

Example:

The Siemens standard cycles are called up using some of the ISO Dialect mode G functions (e.g. G28). DISPLOF is programmed at the start of the cycle, with the result that the ISO Dialect G commands remain active for the display.

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```
PROC  CYCLE328 SAVE DISPLOF
N10 ...
...
N99 RET
```

Procedure:

- External main program calls Siemens shell cycle.
  Siemens mode is selected implicitly on the shell cycle call.
- DISPLOF freezes the block display at the call block;
  the G code display remains in external mode. This display is refreshed while the Siemens cycle is running.

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1.2.2 Zero offset

The zero offsets (ZO) of ISO Dialect mode and the equivalent zero offsets of the Siemens mode are shown in Fig. 1-1.

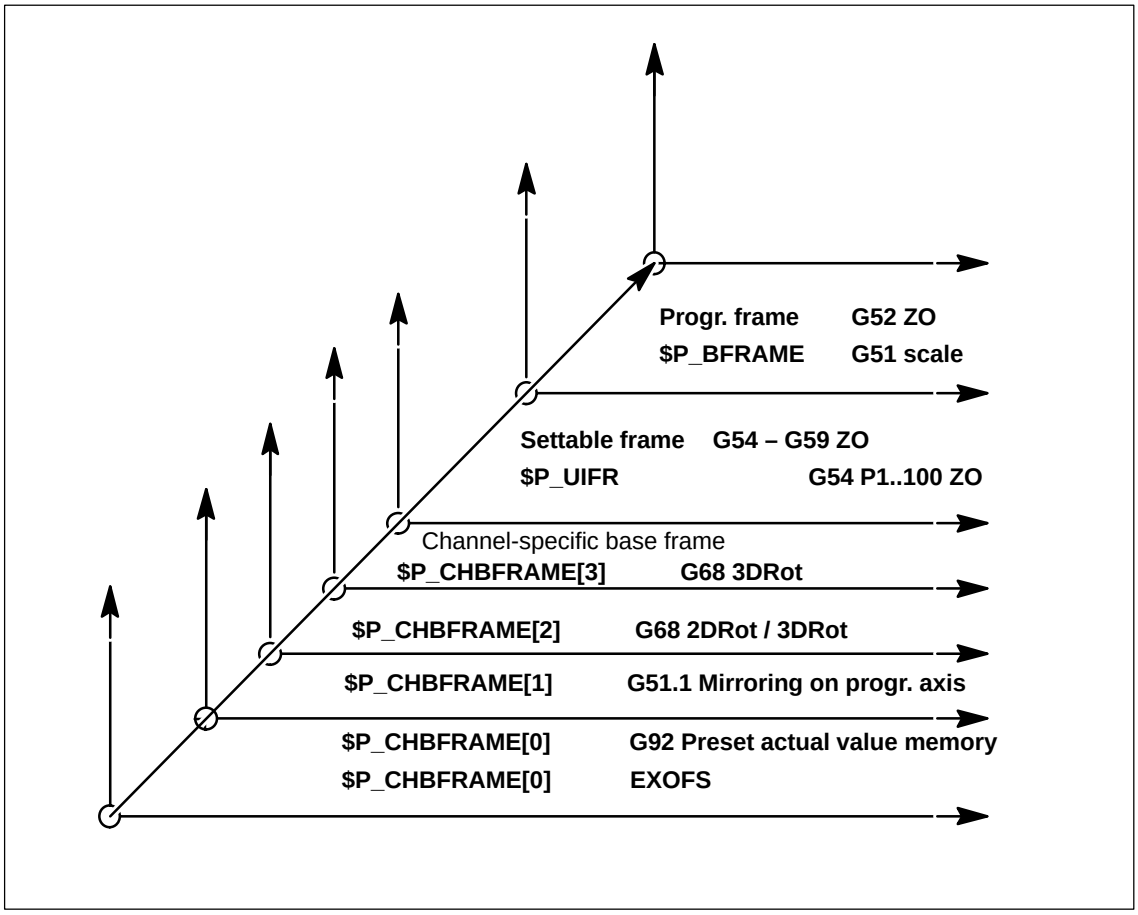


Fig. 1-1 Instantaneous mapping of the ISO functions onto the Siemens frames

The zero offsets that are available in ISO mode are mapped onto the existing Siemens frames. No separate frames exist for ISO mode. Active zero offsets are incorporated in both language modes.
Changes in ISO mode have an immediate effect in Siemens mode and vice–versa.

Zero offsets exist in both ISO Dialect–T and ISO Dialect–M:

- G52 is a programmable, additive ZO that remains active until the end of the program or a reset
- G54 to G59 are settable zero offsets
- G54 P1...P100 are additional settable zero offsets
- G54 P0 is an “external ZO” EXOFS

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1.2.3 Writing a zero offset with G10

G10 can be used to write the zero offsets from the part program.

```
G10 L2 P1...P6 X.. Y..      ;G54.. G59
G10 L20 P1...P100           ;Additional, settable ZO
G10 L2 P0                   ;External ZO EXOFS
```

These zero offsets are mapped onto the same frames as the zero offsets that already exist in ISO Dialect-M.

1.2.4 Decimal point programming

There are two notations for the interpretation of programming values without a decimal point in ISO Dialect mode:

- **Pocket calculator type notation**
  Values without decimal points are interpreted as mm, inch or degrees.
- **Standard notation**
  Values without decimal points are multiplied by a conversion factor.

The setting is defined by MD 10884, see Chapter 4 “Start-up”.

There are two different conversion factors, **IS-B** and **IS-C.** This evaluation refers to addresses X Y Z U V W A B C I J K Q R and F.

Example linear axis in mm:
X 100.5 corresponds to value with decimal point: 100.5 mm
X 1000 pocket calculator type notation: 1000 mm
standard notation: IS-B: 1000* 0.001= 1 mm
IS-C: 1000* 0.0001 = 0.1 mm

Table 1-4 Different conversion factors for IS-B and IS-C

Address	Unit	IS-B	IS-C
Linear axis	mm	0.001	0.0001
Linear axis	inch	0.0001	0.00001
Rotary axis	deg	0.001	0.0001
F feed G94	mm	1	1
F feed G94	inch	0.01	0.01
F feed G95	mm	0.01	0.01
F feed G95	inch	0.0001	0.0001
F lead ISO Dialect-M	mm	0.01	0.01
F lead ISO Dialect-M	inch	0.0001	0.0001
F lead ISO Dialect-T	mm	0.0001	0.0001
F lead ISO Dialect-T	inch	0.000001	0.000001
C chamfer	mm	0.001	0.0001
C chamfer	inch	0.0001	0.00001
R radius, G10 tool compensation	mm	0.001	0.0001

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Table 1-4 Different conversion factors for IS-B and IS-C, continued

Address	Unit	IS-B	IS-C
R radius, G10 tool compensation	inch	0.0001	0.00001
Q ISO Dialect-M	mm	0.001	0.0001
Q ISO Dialect-M	inch	0.0001	0.00001
I, J, K interpolation parameters	mm	0.001	0.0001
I, J, K interpolation parameters	inch	0.0001	0.00001
G04 X or U		0.001	0.001
A contour angle		0.001	0.0001

1.2.5 Scaling and mirroring

Scaling and mirroring are selected with G51.
There are two scaling modes:

- Axial scaling with parameters I, J, K

If I, J, K is not programmed in the G51 block, the default value from the setting data is effective.
Negative axial scaling factors have the additional effect of mirroring.

- Scaling in all axes with scale factor P

If P is not programmed in the G51 block, the default value from the setting data is effective. Negative P values are not possible.
The scale factors are multiplied by either 0.001 or 0.00001.

Note

If a factor other than “1” is programmed for parameters I, J, K or if the address is missing (default value is active for I, J, K), the contour is also scaled.

Example

```
00512 (part program)
N10 G17 G90 G00 X0 Y0          Approach start position
N30 G90 G01 G94 F6000
N32 M98 P0513                  1) Contour as programmed in
                                  subprogram
N34 G51 X0. Y0. I-1000 J1000   2) Mirror contour around X
N36 M98 P0513
N38 G51 X0. Y0. I-1000 J-1000  3) Mirror contour around
                                  X and Y
N40 M98 P0513
N42 G51 X0. Y0. I1000 J-1000   4) Mirror contour around Y
N44 M98 P0513
N46 G50                        Deselect scaling and
                               mirroring
N50 G00 X0 Y0
N60 M30
```

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```
00513 (subprogram)
N10 G90 X10. Y10.
N20 X50
N30 Y50
N40 X10. Y10.
N50 M99
```

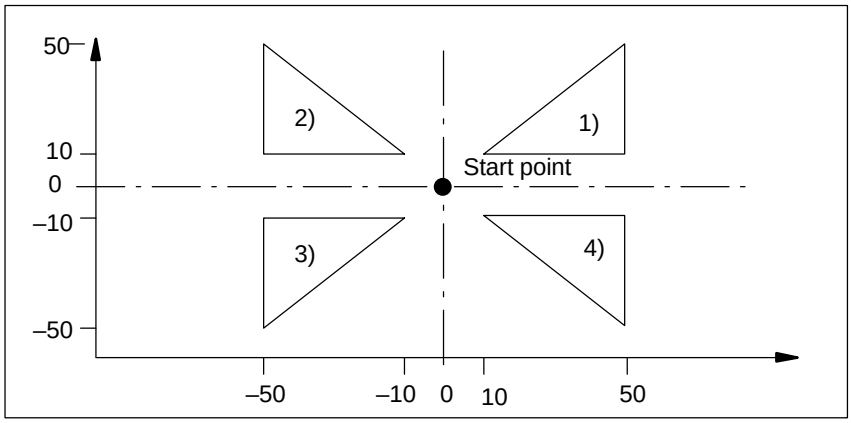


Fig. 1-2 Scaling and mirroring

System parameter settings for the scaling and mirroring example:

MD 22910 $MC_WEIGHTING_FACTOR_FOR_SCALE = 0
MD 22914 $MC_AXES_SCALE_ENABLE = 1
MD 10884 $MN_EXTERN_FLOATINGPOINT_PROG = 0
MD 10886 $MN_EXTERN_INCREMENT_SYSTEM = 0

Axial scaling is not possible when MD $MC_AXES_SCALE_ENABLE = 0.

The reference point during scaling is always the workpiece zero; it is not possible to program a reference point.

Mirroring

Mirroring is selected with G51.1.
Mirroring is performed around a mirror axis that runs parallel to X, Y or Z and whose position is programmed with X, Y or Z. G51.1 X0 is used to mirror about the X-axis and G51.1 X10 is used to mirror about an axis that runs parallel to the X-axis at a distance of 10 mm.

All axes in the channel and not just the geometry axes can be mirrored.
G51.1 functions additively, i.e. following N5 G51.1 X10 and N10 G51.1 Y10, mirroring in X and V is active.

Example

G51.1 X80.
Mirroring is performed around a mirror axis that runs parallel to Y and that crosses the X-axis at position 80.

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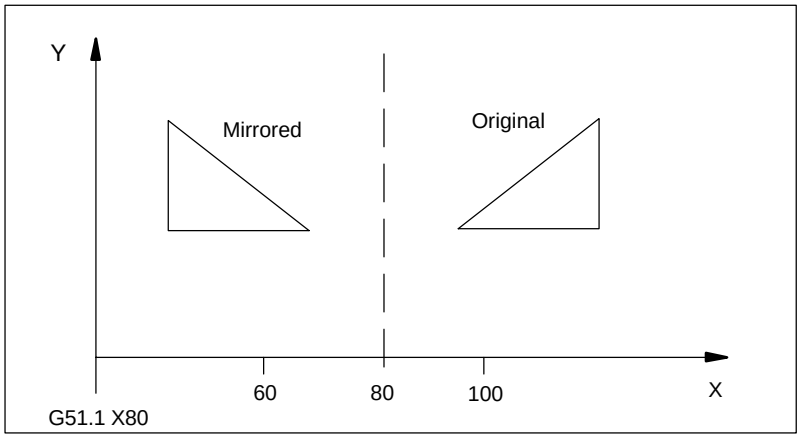


Fig. 1-3 Mirroring around a mirror axis parallel to Y

If the standard notation is active (see Subsection 1.2.4), the axis positions without a decimal point are interpreted in internal units.

Mirroring is deselected with G50.1 X0 Y0. It can also be deselected for individual axes. Following G50.1 X0, mirroring is only deselected for the X-axis; mirroring around all other axes remains active.

G51.1 and G50.1 must be in a block of their own.
G51.1 is mapped onto channel–specific base frame [1]. For this purpose, MD 28081 $MC_MM_NUM_BASE_FRAMES >=2 must be set.
When base frame[1] is changed in Siemens mode, it directly affects the function in ISO mode.
If the frame is deleted in every frame component, this corresponds to a G50.1 X0 Y0.. in all axes.

G51.1 is deselected on a Reset.

1.2.6 2D rotation G68/69 (ISO Dialect-M)

The coordinate system is rotated about the vertical axis of the selected plane.

Programming G68 X.. Y.. R..

X.. Y..:	Coordinates of the pivot point related to the current workpiece zero. If a coordinate is not programmed, the pivot point is taken from the actual value memory. The value is always interpreted as an absolute value.
R:	The angle of rotation is interpreted as an absolute or incremental value depending on G90/G91. If an angle is not programmed, the angle from setting data $SA_DEFAULT_ROT_FACTOR_R is active.
	G68 must be in a block of its own.
G69	Rotation Off; Additional codes can be programmed in this block.

G68 is mapped onto channel-specific base frame 2. For this purpose, machine data $MC_MM_NUM_BASE_FRAMES >= 3 must be set.

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A programmed angle R is not entered in setting data $SA_DEFAULT_ROT_FACTOR_R. This setting data can only be written manually and is effective provided that no R has been programmed in the G68 block.

1.2.7 3D rotation G68/69 (ISO Dialect-M)

G code G68 has been expanded for 3D rotation.

Programming

G68 X.. Y.. Z.. I.. J.. K.. R..

X.. Y.. Z..:	Coordinates of the pivot point related to the current workpiece zero. If a coordinate is not programmed The pivot point is at the workpiece zero. The value is always interpreted as an absolute value. The coordinates of the pivot point act like a zero offset. A G90/91 in the block has no effect on the G68 command.
I.. J.. K..:	Vector in the pivot point. The coordinate system is rotated about this vector by the angle R.
R:	Angle of rotation, always interpreted as an absolute value. If an angle is not programmed, the angle from setting data 42150 $SA_DEFAULT_ROT_FACTOR_R is active.

G68 must be in a block of its own.

The distinction between 2D and 3D rotation is determined solely by programming the vector I, J, K. If no vector exists in the block, G68 2DRot is selected. If a vector exists in the block, G68 3DRot is selected.

If a vector of length 0 (I0, Y0, K0) is programmed, the alarm 12560 “Programmed value lies outside the permissible limits” is output.

With G68, two rotations can be connected in series. If a G68 is not already active in a block containing G68, the rotation is written into channel-specific base frame 2.If G68 is already active, the rotation is written in channel-specific base frame 3. This means that both rotations are activated in sequence.

With G69, 3D rotation is terminated. If two rotations are active, they are both deactivated with G69. G69 does not have to be in a block of its own.

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1.2.8 Polar coordinates (for ISO Dialect-M only)

In ISO Dialect mode, NC program sections programmed with polar coordinates must commence with start command G16. Until the end command G15 is reached, the coordinates of the end points are interpreted as the polar coordinate values for radius and angle in the current plane.

The first axis of the plane is the polar radius, the second axis is the polar angle, i.e. X is the radius and Y is the angle for G17.

After G16 a new pole is set in each block up to G15, with the result as follows for G17:

- G90 X The pole is at the workpiece zero
- G91 X The pole is at the current position
- No X in the block The pole is at the workpiece zero

If the pole is moved from the current position to the workpiece zero, the radius is calculated as the distance from the current position to the workpiece zero.

Example:

```
N5 G17 G90 X0 Y0
N10 G16 X100. Y45.  Polar coordinates ON, pole is
                    workpiece zero, position X 70,711
                    Y 70,711 in Cartesian coordinate
                    system
N15 G91 X100 G90 Y0 Pole is current position, i.e. position
                    X 170,711 Y 70,711
N20 Y90.            No X in block, pole is at
                    workpiece zero,
                    Radius = SORT(X*X +Y*Y) = 184,776
```

The polar radius is always traversed as an absolute distance; the polar angle can be interpreted as an absolute or incremental value.

Programmed angle

In the case of active polar coordinate programming, the programmed angle can be read via the system variable $P_AP.
This variable is inserted in the shell cycle. Before the new pole is set, with incremental programming, the angle must be stored because the angle will be deleted.

Polar programming is terminated by G15. The polar radius is set to 0.

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1.2.9 Interrupt program with M96/M97 (ASUB)

M96

A subprogram can be defined as an interrupt routine with M96 P<Programmnummer>.

This program is started by an external signal. The first high-speed NC input of the 8 inputs available in Siemens mode is always used to start the interrupt routine.

The function is mapped onto Siemens syntax: SETINT(1) <Programmname> [PRIO=1].

M97

M97 is used to suppress starting of the interrupt routine. The interrupt routine can then only be started by the external signal following activation with M96.

This corresponds to Siemens syntax: ENABLE(1).

The M function numbers for the interrupt function are set via machine data. With machine data 10804: $MN_EXTERN_M_NO_SET_INT, the M number is used to activate an interrupt routine and with MD 10806: $MN_EXTERN_M_NO_DISABLE_INT the M number is used to suppress an interrupt routine.

Only non–standard M functions are permitted to be set. M functions M96 and M97 are set as defaults. To activate the function, bit 0 must be set in machine data 10808: $MN_EXTERN_INTERRUPT_BITS_M96. These M functions will not be output to the PLC in this case. If bit 0 is not set, the M functions will be interpreted as conventional auxiliary functions.

On completion of the ”Interrupt” program, the end position of the part program block that follows the interruption block is approached. If processing of the part program has to continue starting from the interruption point, there must be a REPOS instruction at the end of the ”Interrupt” program, e.g. REPOSA. For this purpose the interrupt program must be written in Siemens mode.

The M functions for activating and deactivating an interrupt program must be in a block of their own. If further addresses other than “M” and “P” are programmed in the block, alarm 12080 (syntax error) is output.

Machine data

MD $MN_EXTERN_INTERRUPT_BITS_M96:

Bit 0: = 0, interrupt program is not possible, M96/M97 are conventional M functions
= 1, activation of an interrupt program with M96/M97 is permitted

Bit 1: = 0, execution of part program continues from the final position of the block following the interruption block
= 1, execution of the part program continues from the point of interruption
(evaluated in interrupt program (ASUB), return with/without REPOSL)

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Bit 2: = 0, the interrupt signal interrupts the current block immediately and starts the interrupt routine
= 1, the interrupt routine is not started until the end of the block

Bit 3: = 0, the machining cycle is interrupted on an interrupt signal
= 1, the interrupt program is not started until the end of the cycle
(evaluated in the shell cycles)

Bit 3 must be evaluated in the shell cycles and the cycle sequence must be adapted accordingly.

Bit 1 must be evaluated in the interrupt program. If bit 1 = TRUE, on completion of the program, REPOSL must be used to reposition at the interruption point.

Example:

```
N1000 M96 P1234 ;Activate ASUB 1234.spf On the rising
                ;edge of the first high–speed input,
                ;program 1234.spf is activated
       „
       „
N3000 M97       ;Deactivate the ASUB
```

Rapid lifting (LIFTFAST) is not performed before the interrupt program is called. On the rising flank of the interrupt signal, depending on machine data $MN_EXTERN_INTERRUPT_BITS_M96, the interrupt program is started immediately.

Limitations in Siemens mode

The interrupt routine is handled like a conventional subprogram. This means that in order to execute the interrupt routine, at least one subprogram level must be free. (12 program levels are available in Siemens mode, there are 5 in ISO Dialect mode).

The interrupt routine is only started on a signal transition of the interrupt signal from 0 to 1. If the interrupt signal remains permanently set to 1, the interrupt routine will not be restarted.

Limitations in ISO Dialect mode

One program level is reserved for the interrupt routine so that all permissible program levels can be reserved before the interrupt program is called.

Depending on the machine data, the interrupt program will also be started when the signal is permanently on.

1.2.10 Comments

Parentheses are interpreted as comment characters in ISO Dialect mode. In Siemens mode, “;” is interpreted as a comment. As a simplification, “;” is also interpreted as a comment in ISO dialect mode.

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1.2.11 Skip block

In ISO Dialect mode, a skipped block is identified by “/”. This block is skipped when the relevant skip level is active. A block that is skipped must still be syntactically error-free.

Skip levels /1 to /9, which are possible in ISO Dialect mode, are mapped onto Siemens skip levels /0 to /7.
If the skip character “/” is programmed alone, without a level, level 1 is active by default in ISO mode.

If the skip identifier is in the middle of the block, an alarm is displayed in ISO Dialect mode.

1.2.12 Auxiliary function output

M functions

ISO Dialect mode

M functions are output to the PLC as auxiliary functions. Only M98 and M99 are exceptions. All other predefined M functions are transferred to the PLC as auxiliary functions.
The following are predefined M functions:
M17, M40, M41, M42, M43, M44, M45, M70, M96, M97, M98, M99.

Spindle axis changeover with M29

In ISO Dialect mode, the spindle is switched to axis mode using M29.
The M function number can also be set variably with machine data.
MD 20095 $MC_EXTERN_RIGID_TAPPING_M_NR is used to preset the M function number. The machine data is only effective in external language mode and is initialized with M29. It may only be assigned M function numbers which are not used as default M functions. M function numbers M0–M5, M30, M98, M99 are not allowed.
The same function is executed in Siemens mode with M70.
MD 20094 $MC_SPIND_RIGID_TAPPING_M_NR is used to preset the M function number. The machine data is only effective in Siemens mode and is initialized with M70. This allows an M function other than M70 to be used for the spindle switchover in Siemens mode. The machine data may only be assigned M function numbers which are not used as default M functions. The following are not allowed: M0–M5, M17, M19, M30, M40–M45.

H functions

All H functions are output to the PLC as auxiliary functions with ISO Dialect-M.
In ISO Dialect T, G code system A, H is the incremental distance of the 4th axis provided that a 4th axis exists.

T function

T functions are output to the PLC as auxiliary functions. T has the additional meaning of a tool selection.

D function

The D function is output to the PLC as an auxiliary function. With ISO Dialect-M, tool length compensation is activated with address D.

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B function

If B is not an axis, the B function is output to the PLC with address extension H1=.
Example: B1234 is output as H1=1234.

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1.2.13 Approach 1st reference point

CYCLE328 is called up automatically when ISO Dialect command “G28 <Achse>” is read in. <Achse> specifies the intermediate position (incremental or absolute) via which the reference point is to be approached. The intermediate position and the reference position are then approached in positioning mode.

The cycle is only valid for the axes supported by ISO Dialect:

- ISO Dialect–M: X, Y, Z (A, B, C, U, V, W)
- ISO Dialect–T: X, Z, Y (C)

The cycle always runs with radius programming (DIAMOF). When the cycle is terminated, the G commands that were active before the cycle was called are effective again.

Before the 1st reference point is approached, various machine data must be set, see Chapter 4 “Startup”.

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1.3 Subprogram and macro technology

1.3.1 Subprogram technology

Subprogram calls

Subprogram calls are programmed with M98 in ISO Dialect.
For the program syntax, see Fig. 1-4.

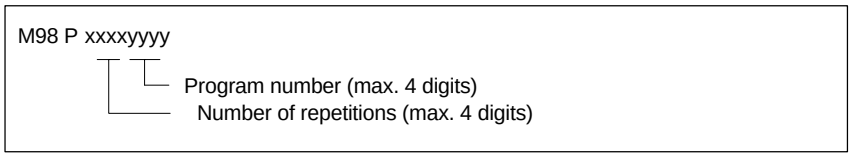


Fig. 1-4 Description of parameters allowed

The program syntax M98 Pxxxxyyyy is used to call a subprogram with the number yyyy and repeat it xxxx times. If the xxxx is not programmed, the subprogram is executed only once. The subprogram name always consists of 4 digits or is extended to 4 digits by adding 0's.
For example, if M98 P21 is programmed, the part program memory is searched for program name 0021.spf and the subprogram is executed once. To execute the subprogram 3 times, program M98 P30021.

SW 6 upwards

Until now the number of program executions (number of repeats) has been programmed in ISO Dialect–M/T in conjunction with the subprogram number at address "P".

As an alternative, the number of subprogram executions can now also be programmed at address "L". The number of the subprogram is still programmed as Pxxxx. If the number of executions is programmed at both addresses, the number of executions programmed at address "L" is valid. The number of subprogram executions lies between 1 and 9999.

Example:

```
N20 M98 P20123      ;Subprogram 123.spf will be executed
                    ;twice
N40 M98 P55 L4      ;Subprogram 55.spf will be executed
                    ;four times
N60 M98 P30077 L2   ;Subprogram 77.spf will be executed
                    ;twice
                    ;The number of executions programmed
                    ;at address "P" =3 is ignored
```

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Subprogram termination

M99 terminates the subprogram.
If M99 Pxxxx is programmed, execution resumes at block number Nxxxx on the return jump to the main program. The block number must always begin with “N”. The system initially searches forwards for the block number (from the subprogram call towards the end of the program). If a matching block number is not found, the part program is then scanned backwards (towards the start of the program).

If M99 appears without a block number (Pxxxx) in a subprogram, the subprogram is terminated and the processor jumps back to the main program to the block following the subprogram call.

If M99 appears without a block number (Pxxxx) in a main program, the processor jumps back to the start of the main program and runs the program again.

These M commands are not output to the PLC.

Subprogram return jump with “RET”

Valid only for ISO Dialect-T.

In the Siemens shell cycles for stock removal (as in ISO Dialect), it is necessary after roughing to resume program execution in the main program after the contour definition. To achieve this, the shell cycle must contain a subprogram return jump to the block after the end of the contour definition. The RET command has been extended with two optional parameters for skipping the blocks with the contour definition in the stock removal cycles after the subprogram call (with G71–G73).

The command RET (STRING: <Satznummer/Label>) is used to resume program execution in the calling program (main program) at the block with <Satznummer/Label>.

If program execution is to be resumed at the next block after <Satznummer/Label>, the 2nd parameter in the RET command must be > 0; RET (<Satznummer/Label>, 1). If a value > 1 is programmed for the 2nd parameter, the subprogram also jumps back to the block after the block with <Satznummer/Label>.

In G70–G73 cycles, the contour to be machined is stored in the main program. The extended RET command is required in order to resume execution after the contour definition in the main program at the end of G70 (finish cut via contour with stock removal cycle). To jump to the next NC block after the contour definition at the end of the shell cycle for G70, the shell cycle must be terminated with the following return syntax:
RET (“N” << $C_Q, 1)

Search direction:

The search direction for <Satznummer/Label> is always forwards first (towards the end of the program) and then backwards (towards the start of the program).

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Example

```
N10 X10. Y20.
N20  G71 P30  Q60 U1 W1 F1000 S1500

    N10 ...                ;Shell cycle for stock
                           removal cycle
    N20 DEF STRING[6]BACK
    N30 ...
    N90
    N100 RET (”N”<<$C_Q, 1) ;Return jump to block after
                           ;Contour def. -> N70


N30 X50. Z20.
N40 X60.
N50 Z55.
N60 X100. Z70.
N70 G70 P30 Q60
N80 G0 X150. Z200.
N90 M30
```

Note

M30 in Siemens mode: is interpreted as a return jump in a subprogram.

M30 in ISO Dialect mode: is also interpreted as the end of the part program in a subprogram.

1.3.2 Siemens language commands in ISO Dialect mode

Certain Siemens language commands are also required in ISO Dialect mode for Shopmill. These commands are executed in ISO Dialect mode. They include subprogram calls with and without transfer parameters (not calls with Lxx, because address L has a different meaning for ISO Dialect), program section repetition and control structures. All other Siemens language commands are denied with an alarm in ISO Dialect mode.

The following Siemens language commands can be programmed when ISO Dialect mode is active:

REPEAT:

REPEAT	Block number> [<Satznummer>] [P..]
REPEAT	UNTIL
REPEATB	<Satznummer> [P..]

Only block numbers, not labels are allowed as start and end identifiers.

IF – ELSE – ENDIF
FOR – ENDFOR
WHILE – ENDWHILE
IF<Bedingung> **– GOTO F**<Bedingung>
CASE

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1.3.3 Extending the subprogram call for contour preparation with CONTPRON

In ISO Dialect, the contour definition for stock removal cycles G70 – G73 is not stored separately in a subprogram (as in SINUMERIK), but appears in the part program (main program). When the cycles are called, the contour definition section is defined by a start and end block number. The cycles receive this block number as a transfer parameter. The indirect subprogram call has been extended for Siemens adaptation cycles.

Previously, subprograms were called indirectly with CALL <Programmname>.

The indirect subprogram call has been extended as follows for access to the contour definition in the main program:

CALL [<Programmname>] BLOCK <Startlabel> TO <Endelabel>

If no program name or an empty string is specified as the program name, i.e. CALL BLOCK <Startlabel> TO <Endelabel>, the search for the program section (start/end label) is made in the program which is currently selected. The search for the labels is also performed in the selected program with MDA, ASUB etc. (i.e. in the case of MDA, the search for the labels is performed not in the MDA buffer but in the program with the selected program name). Programming this syntax directly in a main program has the same effect as repeating a program section in a loop with REPEAT <Startlabel> <Endelabel>, i.e. the search for the start and end label is performed in the program containing the CALL BLOCK ... command.

If a program name is specified, e.g. CALL <progName> BLOCK <Startlabel> TO <Endelabel>, the system searches for the program section (surrounded by the start/end label) in subprogram “progName”.

Example

```
Nxx G71 Pxx Q1110 U.. W..               ;ISO Dialect G
function calls
                      shell cycle CYCLE395.spf
     ;_N_CYCLE395_SPF
     N10 .......
               .......
     Nxxx CYCLE95(....., “N”<<$C_P, “N”<<$C_Q)
                      ;Stock removal cycle with additional
                      ;parameters for start and end label
                      ;label
     PROC CYCLE95(....., STRING[20] startlab, STRING[20]
                      endelab)
     N10 ........
     .........
     Nxxx CONTPRON(...)

     N.... CALL “” BLOCK startlab TO endlab
                      ; Read contour definition or
     N.... CALL BLOCK startlab TO endelab
                      ;call the contour program
     EXECUTE(...)
     ........
     Nxx  M17
Nxxx .....
Nxxx RET (”N”<<$C_Q, 1) ;Return jump to next
```

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```
                          ;block after contour
                          :definition
N1120 ....

Nxxx M30
```

Note

The actual CONTPRON and EXECUTE calls do not have to be modified.

Search for start block number

The start block number (start label) of the contour definition is always searched first towards the end of the program (forwards) and then towards the start of the program (backwards).

If the same block number is programmed more than once, the next contour definition block (label) after the current block in the program is detected as the start of the contour definition (see example). The current block is usually the block in which the stock removal cycle (shell cycle) was called in the main program.

Example

In stock removal cycle CYCLE395, the contour definition which appears between blocks N10–N30 in the main program is to be used (with CALL BLOCK N10 TO N30 in CYCLE395). N40 is the current program line in the main program.

The contour definition block is printed in **bold** lettering in the example.

```
N10 X10. Y20.
N20 X30.
N30 Y10.
N40 G71 P10 Q30... ;Call shell cycle for stock removal cycle
...                ;(In the stock removal cycle
...                ;"CALL BLOCK N10 TO N30" is programmed)
...                ;The lines in bold are detected
...                ;as the contour definition

N50 G90 G54
N60 F1000 G94
N10 X50. Y10.
N20 X33. Y11.
N30 X10.
N50 ....
N..  .....
N800 G71 P10 Q30   ;Call shell cycle for stock removal cycle
...                ;("CALL BLOCK N10 ... ;TO N30" is
                   programmed in the stock removal cycle)
...                ;The lines in italics ;are detected
...                ;as the contour definition

N999 ....
N10 X15.
N20 Y25.
N30 X33.
N1010 ....
N1020 .....
```

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1.3.4 Macro commands with G65, G66 and G67

ISO Dialect

In ISO Dialect mode, macros are called in the part program with G65 Pxx, G66 Pxx. A macro is a set of part program blocks that are terminated with M17.

When the subprogram is called, the mode is switched from ISO mode to Siemens mode.

The following commands are used for selection and deselection:

- G65 Macro call, non-modal
- G66 Macro call, modal
- G67 Deselect modal macro call

Siemens

G commands G65 Pxx and G66 Pxx are used to start macro xx. G65 calls subprogram Pxx once. G66 activates the Pxx subprogram call modally, and the subprogram is then executed in every block containing axis movements (same as MCALL xx). G67 deactivates the modal subprogram call again (equivalent to G80 for cycle calls).

In a part program block with G65 or G66, the address Pxx is interpreted as the program number of the subprogram containing the macro functionality. Address Lxx can be used to define the number of passes of the macro. If a number of passes is not programmed in the calling block, the macro is executed once. All further addresses in this part program block are interpreted (as in ISO Dialect “Macro B”) as transfer parameters, and their programmed values are saved in system variables $C_A–$C_Z. These system variables can be read in the subprograms and evaluated for the macro functionality. If further macros are called with parameters within a macro (subprogram), the transfer parameters must be saved in internal variables in the subprogram before the new macro call.

As in the case of the machining cycles, the language mode is switched implicitly to Siemens mode to allow the definition of internal variables. Therefore, if a further macro call appears in the subprogram, ISO Dialect mode must be selected again first. Because addresses I, J, K can be programmed up to ten times in a block by macro call, an array index must be used to access the system variables for these addresses. The syntax for these three system variables is then $C_I[..], $C_J[..], $C_K[..]. The values are stored in the array in the order programmed. The number of addresses I, J, K programmed in the block is stored in variables $C_I_NUM, $C_J_NUM and $C_K_NUM.

System variables for addresses I, J, K

Addresses I, J, and K can be programmed up to ten times in one block for macro calls with G65 and G66. The system variables $C_x for these three addresses are therefore created in array format. Unlike the rest of the system variables, an array index must always be specified for these three variables. Array index 0 must always be used for cycle calls (e.g. G81);
e.g. N100 R10 = $C_I[0]

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Modal macro calls

With modal macro calls, the programmed addresses are only copied into the system variables in the block containing the call (block with G66). The macro is then executed in every block with an axis movement until it is deselected by G67 or a new macro call is programmed with G66. Only the macro parameters are passed in the block containing the call (= block with G66) for modal macros. The macro is executed for the first time in the next block containing an axis movement. (same procedure as MCALL xx in Siemens mode)

Example of a macro call:

```
_N_M10_MPF:

N10 M3 S1000 F1000
N20 X100. Y50. Z33.
N30 G65 P10 F55 X150. Y100. S2000
N40 X50.
N50 ....
N200 M30
```

Example of a subprogram as macro in Siemens mode:

```
_N_10_SPF:
N10 DEF REAL X_ACHSE, Y_ACHSE, DREHZAHL, VORSCHUB
N15 X_ACHSE = $C_X Y_ACHSE = $C_Y  DREHZAHL = $C_S VORSCHUB
= $C_F
N20 G01 F=VORSCHUB  G95 S=DREHZAHL
...
M17
```

1.3.5 Expansion for macro calls with G65/G66

Until now, automatic switchover to Siemens mode was performed for macro calls with G65/G66.
The user now has the choice whether switchover to Siemens mode takes place when the macro starts or not. Switchover to Siemens mode only takes place when the PROC<Programmname> instruction is used in the first line of the macro program. If this instruction is missing, ISO mode will remain active during execution of the macro program.

The user can therefore decide whether to create local variables (with DEF...) for the purpose of storing transfer variables. It is necessary to switch to Siemens mode to do this using the PROC instruction. The user can also specify that the macro program (e.g. an existing ISO Dialect-M/T macro) is executed in ISO mode.

Example of a macro call:

```
_N_M10_MPF:
N10 M3 S1000 F1000
N20 X100. Y50. Z33.
N30 G65 P10 F55 X150. Y100. S2000
N40 X50.
N50....
N200 M30
```

Example of a subprogram as macro in Siemens mode:

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```
_N_0010_SPF:
PROC 0010     ;Switchover to Siemens mode
N10 DEF REAL X_ACHSE, Y_ACHSE, DREHZAHL, VORSCHUB
N15 X_ACHSE=$C_X Y_ACHSE=$C_Y DREHZAHL=$C_S VORSCHUB=$C_F
N20 G01 F=VORSCHUB G95 S=DREHZAHL
....
N80 M17
```

Example of a subprogram as macro in ISO mode:

```
_N_0010_SPF:
G290          Switchover to Siemens mode

              ;If transfer variables have to be read
N15 X_ACHSE=$C_X Y_ACHSE=$C_Y DREHZAHL=$C_S
N20 G01 F=$C_F G95 S=$C_S
N10 G1 X=$C_X Y=$C_Y
G291          ;switch to ISO mode
N15 M3 G54 T1
N20
....
N80 M99
```

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1.4 Tool change and tool offsets

1.4.1 Tool offsets for ISO Dialect-M

Creating tool data

Since Siemens and ISO Dialect programs are to run alternately on the control, the implementation must use the Siemens tool data memory.
In each offset memory that exists for a tool, the length, geometry and wear in each case are specified.
In Siemens mode, the offset memory is addressed with T (tool number) and D (cutting edge number), or T/D number for short.
In ISO Dialect-M programs, the offset memory is addressed with D (radius) or H (length). This is referred to below as the H number.
In order to establish a unique assignment between this H number and a T/D number, an element $TC_DPH[t,d] has been added to the offset data set.
The H number of the ISO Dialect is entered in this element.

Table 1-5 Example: Tool offset data set

T	D/cutting edge	H number $TC_DPH	Radius	Length
1	1	10		
1	2	11		
1	3	12	100.00	250.00
2	1	13		
2	2	14		
2	3	15		

Example:

Siemens program	ISO Dialect program
N5 T1	N5 T1
N10 G41 D3	N10 G41 D12

When **the H value is programmed** in the ISO Dialect program, the system searches for and activates the matching $TC_DPH after the correction block.

If the correction block does not contain an H number, the compensation cannot be activated in ISO Dialect mode. The H number must be unique.

If an H is programmed but a correction block with the corresponding H number is not found or the associated tool T is not selected, this results in an alarm.

Changing the offset memory

Existing tool offsets can be overwritten with G10. New tool offsets are not created by G10.

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Tool length compensation, geometry:	G10 L10 Pxx Ryy
Tool length compensation, wear:	G10 L11 Pxx Ryy
Tool radius compensation, geometry:	G10 L12 Pxx Ryy
Tool radius compensation, wear:	G10 L13 Pxx Ryy

P specifies the number of the compensation memory and R specifies the value. L1 can be programmed instead of L11.

Setting data $SC_TOOL_LENGTH_CONST must contain the value 17 for the assignment of tool length offsets to geometry axes to be independent of the plane selection. Length 1 is then always assigned to the Z axis.

Selecting the tool length

The tool length and the tool radius are always programmed with D or H.

Example:

T	D/cutting edge	H number $TC_DPH	Radius	Length
2	3	4	10	15

ISO Dialect-M:
T2
G43 H4 or D4 ;Length selection
G42 D4 ;Radius selection

The offset value must be entered twice for ISO Dialect programs which are programmed with different D and H numbers.

Example:

T	D/cutting edge	H number $TC_DPH	Radius	Length
2	3	4	10	15
2	4	5	10	15

ISO Dialect–M:
T2
G43 H4 ;Length selection $TC_DPH4
G42 D5 ;Radius and length selection from $TC_DPH5

Flat D number

If flat D numbers are active, the T is programmed independently of the H number. It is no longer checked that the H number is compatible with the selected tool.

An H number must be assigned to every offset memory, even with flat D numbers.

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Tool management

If tool management is active, replacement tools have the same H number. Duplo numbers are used in order to differentiate.
Offset D1 is activated for the currently selected tool on H99 with active **tool management**.

In ISO Dialect-M, only numerical expressions are permitted as tool identifiers. Strings are no longer permitted as identifiers.
Example: T = ”2”, selection with T2.

Tool length compensation in multiple axes

Tool length offsets can be activated on multiple axes. However, it is not possible to display the resulting tool length compensation.

The Siemens T and D numbers appear in the display for active T and D numbers.
New OPI variables which can be displayed are available for the active ISO Dialect H and D number.

Machine data 22220 $MC_AUXFU_T_SYNC_TYPE is used to define whether the output to PLC takes place during or after the movement.
Machine data 20110 $MC_RESET_MODE_MASK, bit 6 can be used to activate tool length compensation beyond a reset.

Example

Tool selection in ISO Dialect-M:

```
;(Fanuc 0 M tool offset with T, cutting edge number
;(the offsets are written)
;(with G10)

G290

;Tool offset memory T2 cutting edge 1:
N5000 $TC_DP1[2,1]=10   ;Type
N5000 $ TC_DP1[2,1]=7   ;Fanuc H number

;Tool offset memory T3 cutting edge 2:
N5000 $TC_DP1[3,2]=10   ;Type
N5000 $TC_DP1[3,2]=3    ;Fanuc H number

;Tool offset memory T4 cutting edge 3:
N5000 $TC_DP1[4,3]=10   ;Typ
N5000 $TC_DP1[4,3]=4    Fanuc H number

G291

;Write tool offsets
;------------------------
;T2 cutting edge 1
G10 L10 P7 R10
G10 L12 P7 R5

;T3 cutting edge 2
G10 L10 P3 R15
G10 L12 P3 R10

;T4 cutting edge 3
G10 L10 P4 R20
```

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```
G10 L12 P4 R15

N8   G01 G40 F5000 X0 Y0 Z0
N10  X50.
N15  50
N17  Z10.
N20  X0
N25  Y0
N30  X–10 Y-10

N30  T2              ;Tool 2
N33  G43 H7 Z0       ;H number 7
N35  G41 X0 Y0 Z0 D7
N40  X50.
N45  Y50.
N48  Z10.
N50  X0
N55  Y0
N60  G40 X-10 Y-10

N65  T3
N68  G43 H3 Z0
N70  G42 X0 Y0 Z0 D3
N75  X50.
N77  Y50.
N78  Z10.
N80  X0
N85  Y0
N90  G40 X-10 Y-10

N95  T4
N98  G43 H4 Z0
N100 G41 X0 Y0 Z0 D4
N105 X50.
N110 Y50.
N112 Z10.
N115 X0
N120 Y0
N125 G40 X-10 Y-10

M30
```

Machine data 20382 $MC_TOOL_CORR_MOVE_MODE defines whether the compensation is applied in the block containing the selection or the next time the axis is programmed.

Machine data 20270 $MC_CUTTING_EDGE_DEFAULT = 0 defines that no tool length compensation is active initially on a tool change.

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1.4.2 Tool compensation for ISO Dialect-T

The tool data are stored in the Siemens tool data memory.
Each tool consists of four entries, one each for the X and Z axes, radius and tool point direction. The range of values for length and radius compensation is ± 999.999 mm. The value range 0–9 is used for the tool point direction, where 0 and 9 are identical.
The meaning is equivalent to the tool point direction on Siemens turning tools.

You can use either flat D numbers or the conventional tool compensation memory management.

Tool management can also be used.

Selection is made by means of T, whereby T contains the tool number and the cutting edge number. The H number of ISO Dialect-M does not exist.

Txxxxyyyy: xxxx = tool number, yyyy = cutting edge number

Machine data 10890 $MN_EXTERN_TOOLPROG_MODE, Bit 0 is used to define how the T value is interpreted.

Machine data 10888 $MN_EXTERN_DIGITS_TOOL_NO is used to specify the number of digits for the tool number. The digits are counted from the left. The subsequent digits specify the offset number.

If bit 0=1 in MD 10890, the cutting edge number is set equal to the tool number.

Example:

```
$MN_EXTERN_TOOLPROG_MODE=0
$MN_EXTERN_DIGITS_TOOL_NO=2
T1234           ;Auxiliary function T1234 to PLC
                ;Tool number 12
                ;Offset selection D34

T123            ;Auxiliary function T123 to PLC
                ;Tool number 12
                ;Offset selection D3

$MN_EXTERN_TOOLPROG_MODE, Bit 0=1
T12             ;Auxiliary function T12 to PLC
                ;Tool number 12
                ;Offset selection 12
```

Machine data 20382 $MC_TOOL_CORR_MOVE_MODE defines when the compensation is applied: In the block containing the selection or the next time the axis is programmed.

MD 20110 $MC_RESET_MODE_MASK, bit 6 determines whether or not the tool compensation remains active after a reset.

MD 20360 $MC_TOOL_PARAMETER_DEF_MASK, bit 0 can be used to define that the wear value of the transverse axis is calculated as a diameter value. The geometry offset is always interpreted as a radius value.

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Example

Tool selection in ISO Dialect-T:

```
G290
N5000  ;Definition of the tool offset memory D1:
N5000  $TC_DP1[1,1]=10        ;Type
N5000  $TC_DP2[1,1]=9         ;Tool point direction
N5000  $TC_DP6[1,1]=5         ;Radius

N5000  ;Definition of the tool offset memory D2:
N5000  $TC_DP1[2,1]=10        ;Typ
N5000  $TC_DP2[2,1]=1         ;Tool point direction
N5000  $TC_DP6[2,1]=5         ;Radius

G291
;Write tool offset data
N3     G10 P1 X10 Z20 Y30
N5     G10 P2 X20 Y20 Z100

N10    G00 G18 X0 Y0 Z0
N10 T0101                     ;Tool 1, cutting edge 1
N15    G00 X10 Y10 Z10
N20    T0201                  ;Tool 1, cutting edge 1
N25    G00 X10 Y10 Z10
...
M30
```

Changing the offset memory

Existing tool offsets can be overwritten with G10. New tool offsets are not created by G10.

```
G10 P<100 / 10000 X Y R Q       Geometry
G10 P>100 / 10000 X Y R Q       Wear

P100 / 10000    ;MD EXTERN_FUNCTION_MASK, Bit 1 is used to select
                ;whether with P<100 or 10000 geometry or wear
                ;is applicable.
X Y Z           ;Offset values absolute or incremental, depending on
                ;G90/91
U V W           ;Incremental offset values
R               ;Radius
Q               ;Tool point direction
```

1.4.3 Tool-changing cycle

ISO Dialect mode

MD 10717 T_NO_FCT_CYCLE_NAME is used to assign a subprogram to the T command. Every block that contains a T command is executed and the subprogram is subsequently called up. The T value is not output; the T command must be programmed again in the cycle.

System variable $C_T_PROG or $C_D_PROG can be used in the subprogram to check whether the T or D command was programmed. The values can be read out with system variable $C_T or $C_D. If another T command is programmed in the subprogram, no substitution takes place, but the T word is output to the PLC.

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The machine data 10715 M_NO_FCT_CYCLE and
10716 M_NO_FCT_CYCLE_NAME can be used to assign a subprogram to an M function (e.g. M06).

The mapping of M and T programming onto cycle calls has the same effect in ISO Dialect mode as in Siemens mode.

If T and M6 are programmed in the same block, the programmed T number can be scanned with $P_TOOL in the cycle called by M6. The M/T call is also mapped onto the cycle call in the block search. The start of the change cycle after the end of the search run must be initiated by the PLC.

Procedure:

```
N20 T1234
N30 M6        ;Change tool
N40 H3  G43   ;Activate tool length compensation in T1234
N50 T333      ;Tool preselection
N60 G1  X10   ;Offset T1234 is active
N70 M6        ;Load tool 333, D0 H0 active
N80 H4        ;Activate new tool length compensation
N90 .....
```

■

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Notes

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Cycles and Contour Definition 2

2.1 Calling cycles in the ext. CNC syst. using G commands

General description

The functionality of the ISO Dialect cycles is implemented in the standard Siemens cycles:
A shell cycle is called from the ISO Dialect program. All addresses programmed in the block are passed to this shell cycle in the form of system variables. The shell cycle matches the data to the standard Siemens cycle and calls it by name.
Machine manufacturers can replace these shell cycles with their own cycles.

Cycle parameters

Various cycle parameters in channel-specific GUD (Global User Data) must be initialized for the machining cycles. The names and meanings of the GUD are listed in Section 3.2.

Procedure for cycle call via G command

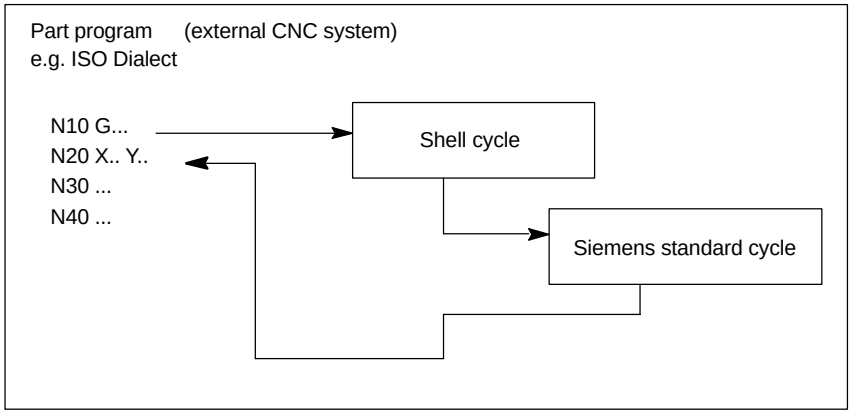


Fig. 2-1 General cycle call in ISO Dialect mode

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Shell cycle

The modifications required due to the ISO Dialect programming syntax are made in the shell cycle. This means that the existing SINUMERIK cycles do not have to be changed. The name of the shell cycle is permanently defined.

Procedure:

1. The cycle (e.g. G81) is programmed in ISO Dialect mode
2. Siemens mode is activated automatically and the associated shell cycle is called (see Fig. 2-2)
3. The shell cycle calls the associated Siemens standard cycle

It is not necessary to program G290. The external CNC system is automatically activated on the return jump.

!

Important

The cycles must only be called with G commands.
This ensures that the appropriate cycle parameters are passed to the shell cycle.

The shell cycle **must not** be activated directly with CALL CYCLE3xx!

Modal cycles

If a modal cycle is active, the shell cycle is called in every NC block. If no axis positions (X, Y or Z) are programmed in the NC block, the Siemens standard cycle is not called.

Addresses programmed in the block (F etc.) are activated via the shell cycle. If no feedrate was programmed, for example, the current feedrate is used as the path feed.

Cycle parameters can be programmed in the following blocks while a modal cycle is active. These parameters are copied into the system variables so that the shell cycle uses the modified parameters.
Modal cycles are, in contrast to modal macros, already executed in the calling block (e.g. block with G81 etc.).

Deselecting the cycle:
Deselection is performed with G80 or with a function of the 1st G group.

Example:

```
N10 G81 X10. Y20. Z–15. R5 F1000
                        Drilling position X10mm, Y20mm
                        Drilling depth Z-15mm
                        Reference plane 5mm
                        Drilling feed F.. (mm/min ormm/rev)
N20 X50. Y30. R10       Drilling position X50mm, Y30mm,
                        New reference plane 10mm
N30 G80                 Delete cycle G81
```

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2.2 Global user data (GUD)

Table 2-1 GUD7 for programmed cycle values (ISO Dialect program data)

GUD	Description/use	CYCLE
Real values		
_ZFPR[0]	Initial plane (current position on 1st call with G..), retraction position active on G98	381M, 383M, 384M, 387M
_ZFPR[1]	Reference plane, retraction position active on G99 (retraction only possible to initial position with G87).	381M, 383M, 384M, 387M
_ZFPR[2]	Final drilling depth, absolute	381M, 383M, 384M, 387M
_ZFPR[3]	Retraction position, depending on G98/G99 (initial plane/R plane)	381M, 383M, 384M, 387M
_ZFPR[4]	Drilling feed	381M, 383M, 384M, 387M
_ZFPR[5]	Dwell (s) at final depth (G82/G89/G76/G87)	381M, 384M, 387M
_ZFPR[6]	1st drilling depth (single drilling depth) incr. (G73/G83)	383M
_ZFPR[7]	1st drilling depth, absolute (G73/G83)	383M
_ZFPR[8]	Lift-off/infeed distance (G76)	387M
_ZFPR[9]	Speed for tapping (G74/G84)	384M
_ZFPR[20]	Initial plane (current position on 1st call)	383T, 384T, 385T
_ZFPR[21]	R plane	383T, 384T, 385T
_ZFPR[22]	Final drilling depth, absolute	383T, 384T, 385T
_ZFPR[23]	Retraction position (1=G98, 2=G99)	383T, 384T, 385T
_ZFPR[24]	Thread pitch/drilling feed	376T, 383T, 384T, 385T
_ZFPR[25]	Dwell time at final depth	383T, 384T, 385T
_ZFPR[26]	Speed for tapping	384T
_ZFPR[27]	End point X	371T, 372T, 373T, 376T
_ZFPR[28]	End point Z	371T, 372T, 373T, 376T
_ZFPR[29]	Start point offset X (taper thread)	371T, 372T, 376T
_ZFPR[30]	Thread start point X	376T
_ZFPR[31]	Thread start point Z	376T
_ZFPR[32]	First drilling depth	383T
Integer values		
_ZFPI[0]	Current G code of ISO Dialect drilling cycle	381M, 383M, 384M
_ZFPI[1]	M function for spindle start (M3, M4) after spindle stop	381M, 384M
_ZFPI[20]	Current G code of threading cycle/drilling cycle	383T, 384T, 385T

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Table 2-1 GUD7 for programmed cycle values (ISO Dialect program data), continued

GUD	Description/use	CYCLE
_ZFPI[21]	Spindle direction (3=M3, 4=M4)	383T, 384T, 385T
_ZFPI[22]	Stock removal mode Roughing	370T, 371T, 372T, 373T
_ZFPI[23]	Machining mode Deep hole/Drilling	383T

Table 2-2 GUD7 for cycle setting data (ISO Dialect setting data)

GUD	Description/use	
Real values		
_ZSFR[0]	Safety clearance to reference plane	381M, 383M
_ZSFR[1]	Retraction amount for chipbreaking (G73)	383M
_ZSFR[2]	Angle offset for oriented spindle stop, tool must be oriented in +X direction (G76) Retraction direction: –X G17 plane XY –Z G18 plane ZX –Y G19 plane YZ	387M
_ZSFR[20]	Safety clearance to reference plane	383T, 384T
_ZSFR[21]	Safety clearance to chip break	383T, 385T
Integer values		
_ZSFI[0]	0 =Drilling axis is perpendicular to plane (default) 1 =Drilling axis always “Z”	381M, 383M, 384M,387M
_ZSFI[1]	0 = Rigid tapping 1 = Tapping with compensating chuck 2 = Deep hole tapping with chipbreaking 3 = Deep hole tapping with chip removal	384M,387M
_ZSFI[2]	Retraction speed factor (1–200%) for tapping (G74/G84)	384M
_ZSFI[3]	Polar coordinates 0 = OFF 1 = ON	381M, 383M, 384M,387M
_ZSFI[20]	Deep hole drilling with chip breaking/removal	383T, 385T
_ZSFI[22]	Factor for retraction speed	384T
_ZSFI[23]	Dwell time with G95, 0 = seconds, 1 = revolution	383T
_ZSFI[24]	Number of noncuts	376T
_ZSFI[25]	Cutting edge angle	376T
_ZSFI[26]	Thread run-out distance (n*pitch)	376T
_ZSFI[27]	Min. infeed depth	376T
_ZSFI[28]	Final machining allowance	376T
_ZSFI[29]	Distance traversed for grooving cycle	374T
_ZSFI[30]	Cutting depth for stock removal cycle	371T, 372T
_ZSFI[31]	Distance traversed for stock removal cycle	371T, 372T
_ZSFI[32]	X-axis infeed value for contour repetition	373T
_ZSFI[33]	Z-axis infeed value for contour repetition	373T

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Table 2-2 GUD7 for cycle setting data (ISO Dialect setting data), continued, continued

GUD	Description/use	
_ZSFI[34]	Number of divisions for contour repetition	373T
_ZSFI[39]	G code system 2 = B, 1 = A, 3 = C	300, 328, 330, 370T, 371T, 372T, 373T, 374T, 376T

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2.3 Drilling cycles (ISO Dialect-M)

2.3.1 Overview and parameter description

The drilling cycles are modal. While a drilling mode is active, you only have to program the new parameters in order to make parameter modifications.

There is **no** traversing movement if:

- No value is programmed in the NC block for X, Y and Z
- The number of repetitions K = 0 was programmed

The retraction position is valid for all drilling cycles

- G98 Retraction to initial plane
- G99 Retraction to reference plane

Overview

Table 2-3 Overview of drilling cycles

External cycle call	Description
G73 X.. Y.. Z.. R.. F.. Q..	Deep hole drilling cycle with chipbreaking
G74 X.. Y.. Z.. R.. F.. P..	Counterclockwise tapping cycle
G76 X.. Y.. Z.. R.. F.. Q.. P..	Fine drilling cycle
G80	Cycle off; the cycle is also deselected by programming a G function of the 1st G group.
G81 X.. Y.. Z.. R.. F..	Drilling cycle; drilling, retraction with G00
G82 X.. Y.. Z.. R.. F.. P..	Drilling cycle; drilling, dwell, retraction with G00
G83 X.. Y.. Z.. R.. F.. Q..	Deep hole drilling cycle with chip removal
G84 X.. Y.. Z.. R.. F.. P..	Clockwise tapping cycle
G85 X.. Y.. Z.. R.. F..	Drilling cycle; drilling, retraction with drilling feed
G86 X.. Y.. Z.. F.. R.. K..	Drilling cycle, retraction with G00
G87 X.. Y.. Z.. F.. R.. P.. Q.. K..	Reverse countersinking
G89 X.. Y.. Z.. F.. R.. P.. K..	Drilling cycle, retraction with machining feed

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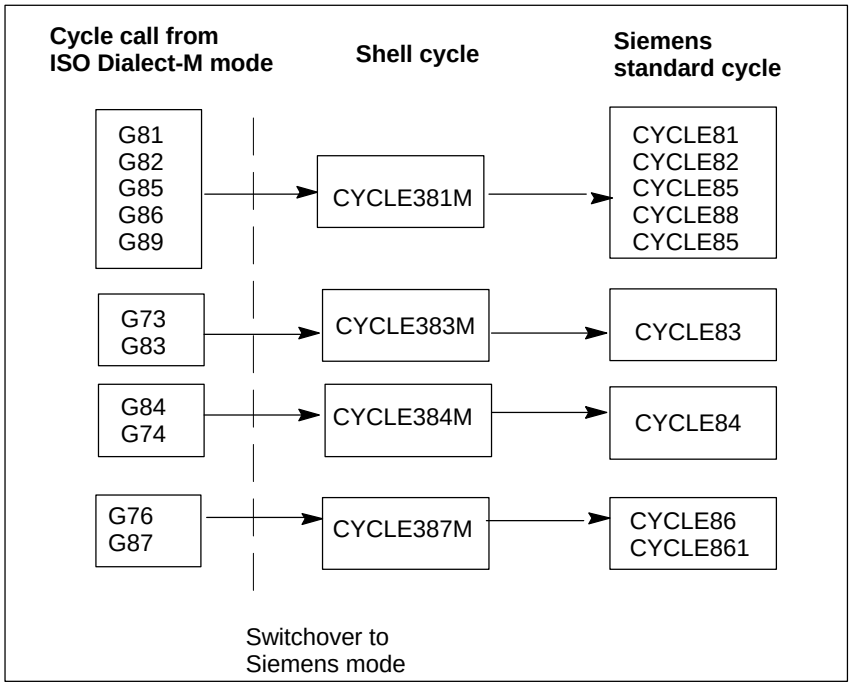


Fig. 2-2 Assignment of the cycle call in ISO Dialect-M mode via shell cycle for Siemens standard cycle

Example: ISO Dialect-M

```
N10 G81 X100. Z–50. R20  F100
```

G81 automatically calls the shell cycle CYCLE381M.
The calculations are performed in the shell cycle and the standard drilling cycle CYCLE81 is then called.

Parameter description

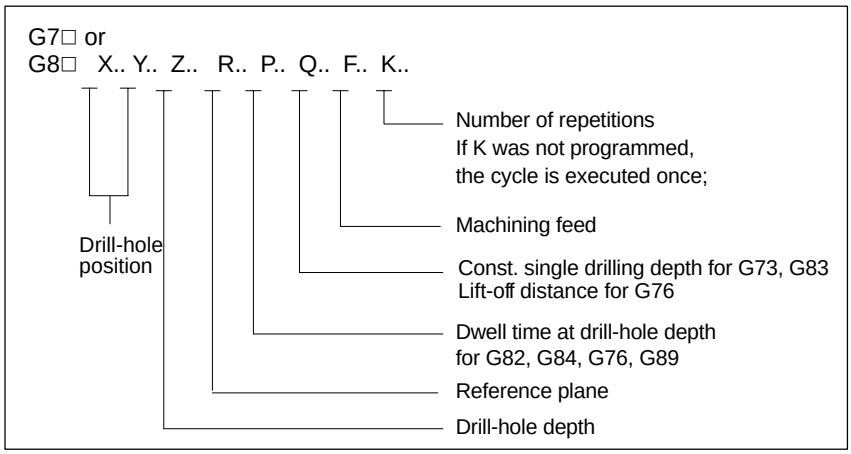


Fig. 2-3 Description of parameters allowed for G17 (X/Y plane)

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Plane

Table 2-4 Definition of the plane

Defined plane	Position of hole	Depth
G17	X, Y	Z
G18	Z, X	Y
G19	Y, Z	X

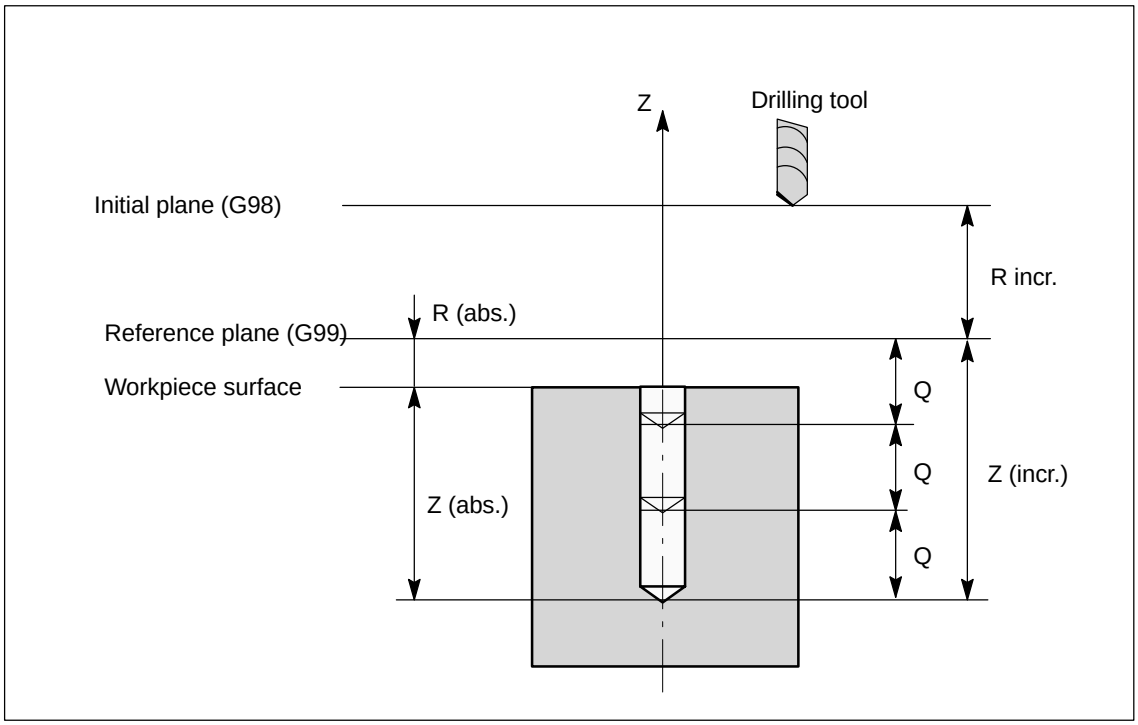


Fig. 2-4 Example of deep hole drilling cycle G83 with defined plane G17; representation of initial plane, reference plane and parameters

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2.3.2 Description of shell cycle CYCLE381M

In ISO Dialect-M mode, the call is performed with G commands G81, G82, G85.

Notes

The drilling axis must be defined via GUD _ZSFI[0] (see Section 2.2).

The direction from initial plane to reference plane must be identical to the direction from reference plane to final depth.

GUD _ZSFR[0] can be used to enter a safety clearance. If the safety clearance was already allowed for when programming the reference plane, the value 0 must be entered in _ZSFR[0].

If no reference plane was programmed, the drilling is performed starting on the initial plane (current position).

The final drilling depth must be programmed. Otherwise, an alarm is displayed.

If no feedrate is programmed, the current feedrate is used as the drilling feed.

Note

Alarms are listed with their alarm number and description in Chapter “Alarms”.

2.3.3 Description of shell cycle CYCLE383M

In ISO Dialect-M mode, the call is performed with G commands G73, G83.

Notes

The drilling axis must be defined via GUD _ZSFI[0] (see Section 2.2).

The direction from initial plane to reference plane must be identical to the direction from reference plane to final depth.

GUD _ZSFR[0] can be used to enter a safety clearance. If the safety clearance was already allowed for when programming the reference plane, the value 0 must be entered in _ZSFR[0].

If no reference plane was programmed, the drilling is performed (in steps) starting on the initial plane (current position).

The final drilling depth and the single drilling depth Q must be programmed. Otherwise, an alarm is displayed.

If no feedrate is programmed, the current feedrate is used as the drilling feed.

Note

Alarms are listed with their alarm number and description in Chapter “Alarms”.

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Deep hole drilling with chip removal

The safety clearance in the standard Siemens cycle is determined by a formula, i.e. it cannot be selected by the user.

For stock removal, retraction is to the reference plane.

Single drilling depth “Q”:

- If “Q” is missing or Q $\leq$ 0, an alarm is displayed.
- If Q > total depth, one drilling operation is executed up to the final depth.
- If Q > total depth/2, the 1st drilling operation is performed with the value of Q. The remainder is then drilled in one drilling operation.
- If Q < total depth/2, the single depth is machined until the remaining depth < Q/2. The remainder is then subdivided into 2 infeed movements of the same size.

Deep hole drilling with chipbreaking

The amount of retraction for chipbreaking is defined via GUD _ZSFR[1].

- _ZSFR[1] > 0 Retraction amount as entered
- _ZSFR[1] $\leq$ 0 Retraction amount is always 1mm

Single drilling depth “Q”:

- If “Q” is missing or Q $\leq$ 0, an alarm is displayed.
- If Q > total depth, one drilling operation is executed up to the final depth.
- If Q < total depth, the single drilling depth is machined until the remaining depth $\leq$ Q. The remainder is then machined in one drilling operation.

2.3.4 Description of shell cycle CYCLE384M

In ISO Dialect-M mode, the call is performed with G commands G74, G84.

Notes

The drilling axis must be defined via GUD _ZSFI[0] (see Section 2.2).

The direction from initial plane to reference plane must be identical to the direction from reference plane to final depth.

GUD _ZSFR[0] can be used to enter a safety clearance. If the safety clearance was already allowed for when programming the reference plane, the value 0 must be entered in _ZSFR[0].

The speed of rotation during retraction can be controlled via GUD _ZSFI[2]. Example: _ZSFI[2]=80, the retraction takes place with 80 % of the drilling speed.

If no reference plane was programmed, the drilling is performed starting on the initial plane (current position).

The final drilling depth must be programmed. Otherwise, an alarm is displayed.

If no feedrate is programmed, the current feedrate is used as the drilling feed.

If the drilling feed is specified in mm/min (inch/min), the programmed feed value is converted to the appropriate revolutional feedrate, depending on the speed last programmed, and passed to the standard tapping cycle CYCLE84 as a lead value.

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Note

Alarms are listed with their alarm number and description in Chapter “Alarms”.

2.3.5 Description of shell cycle CYCLE387M

In ISO Dialect-M mode, the call is performed with G commands G76 and G87.

Notes

The drilling axis must be defined via GUD _ZSFI[0] (see Section 2.2).

The direction from initial plane to reference plane must be identical to the direction from reference plane to final depth.

GUD _ZSFR[0] can be used to enter a safety clearance. If the safety clearance was already allowed for when programming the reference plane, the value 0 must be entered in _ZSFR[0].

The lift-off path is always with reference to the negative direction of the first geometry axis

For plane G17	Lift-off path in –X
For plane G18	Lift-off path in –Z
For plane G19	Lift-off path in –Y

The angle must be therefore be entered such that the tool tip points in the positive direction (+) in the defined plane after the spindle stop.
Example: If plane G17 is active, the tool tip must point in the +X direction.

If no reference plane was programmed, the drilling is performed starting on the initial plane (current position).

The final drilling depth must be programmed. Otherwise, an alarm is displayed.

If no feedrate is programmed, the current feedrate is used as the drilling feed.

If no lift-off amount is programmed, Q = 0 is set. In this case, the cycle is executed without lift-off.

After retracting to the return plane, the tool is moved back to the center of the drill-hole and the spindle is started in the direction of rotation for machining.

Note

Alarms are listed with their alarm number and description in Chapter “Alarms”.

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2.4 Turning and drilling cycles (ISO Dialect-T)

2.4.1 Turning cycles G70 to G76

Table 2-5 Overview of turning cycles

G command	Description
G70	Finishing cycle
G71	Stock removal cycle longitudinal axis
G72	Stock removal cycle transverse axis
G73	Repeat contour
G74	Deep hole drilling and recessing in longitudinal axis (Z)
G75	Deep hole drilling and recessing in facing axis (X)
G76	Multiple thread cutting cycle

The cycle parameters for G71 to G76 can comprise two G commands. Depending on the addresses programmed in the block, the values of the addresses programmed in the NC block are only saved for use in a subsequent cycle call, or the actual cycle is started. The evaluation of the two cases is performed within the shell cycle.

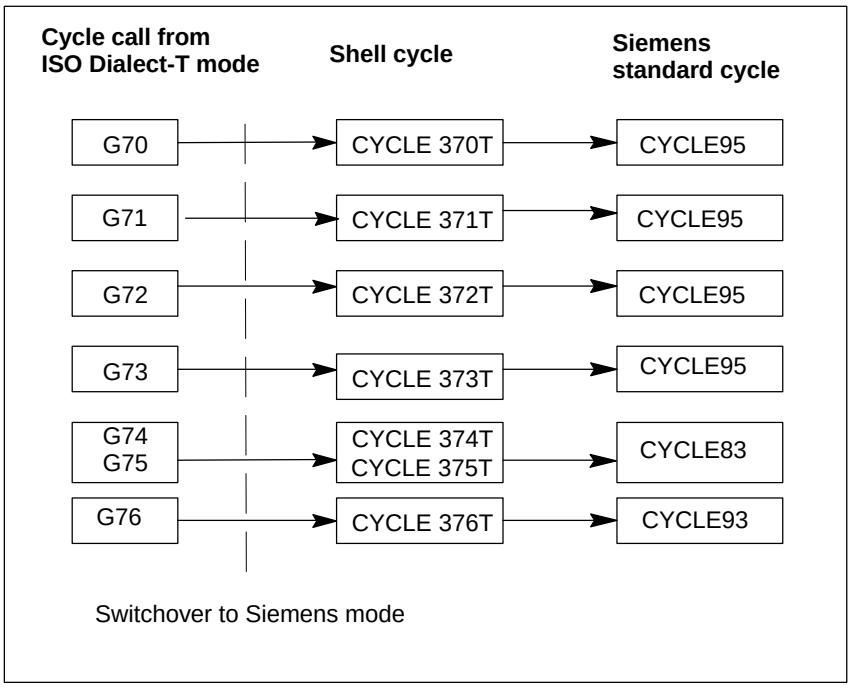


Fig. 2-5 Assignment of the cycle call in ISO Dialect-T mode via shell cycle for Siemens standard cycle

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Finishing cycle G70

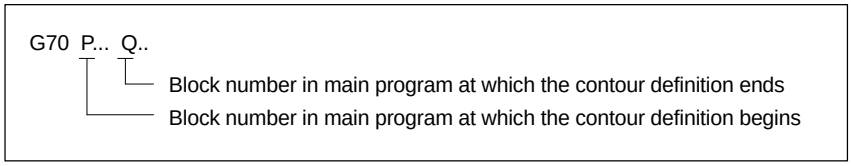


Fig. 2-6 Description of parameters allowed

In ISO Dialect mode, the contour in the main program is not skipped with G70. The program is always resumed at the next part program block following the cycle call.

Example:

When the cycle is called, the contour in N20–N50 is traversed; the part program continues running at N20 after the end of the cycle. G70 is naturally always called up after the contour definition.

```
N10 G70 P20 Q50
N20 X100. Z50.
N30 X200.
N40 Z100.
N50 X250. Z111.
N60 M30
```

Blocks N20–N50 are executed once by the finishing cycle and again by the normal program run.

Stock removal cycle, longitudinal axis G71

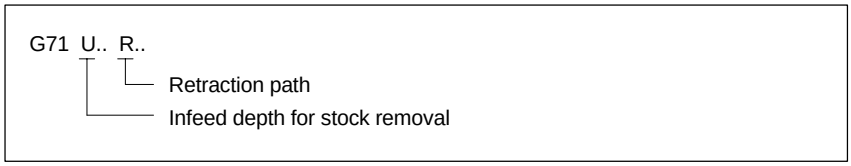


Fig. 2-7 Description of parameters allowed; saving values in GUD

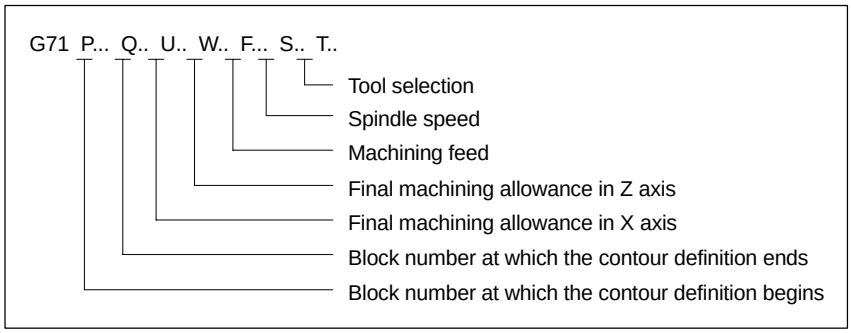


Fig. 2-8 Description of parameters allowed; running cycles

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With G71 in ISO Dialect-T, the contour is followed with the final machining allowance in the main program after stock removal (to remove any remaining straight edges). The program is always resumed at the part program block following the last block of the contour definition. Part program blocks between the cycle call and the first block of the contour description are not executed.

```
N10 G71 P50 Q80 U4 W3 F1000 ...
N20 G1 F0.5 G95 S1000
N30 X30. Z10.
N40 M30
N50 X100. Z50.
N60 X200.
N70 Z100.
N80 X250. Z111.
N90 M30
```

Blocks N20–N40 are skipped and are not executed. If G71 is called in the part program after the contour definition, the program goes into an infinite loop. Allowance must be made for the different continuation patterns of the part program in the shell cycles.

Example: G71 U6 R5 Saving values in GUD

In the example, the values programmed in the shell cycle are read from the system variables ($C_xx) and stored in channel-specific GUD. Separate GUD are available for each cycle (G71–G76); the programmed values therefore remain valid until the next time they are programmed (in an NC block with G71–G76). Case 1 does not have to be programmed, therefore it is advisable to initialize the assigned GUD.

Example: G71 P30 Q50 U3 Starting cycle execution

In the example, the programmed values are only saved in system variables ($C_xx). The system variables are overwritten in every NC block containing a G function for a cycle call. With G71–G73, the cycle always starts running on the G command after which the “P” and “Q” are programmed. With G74–G76, the cycle starts running on the G command after which the X/U or Z/W addresses are programmed. The F, S and T commands in the call line of the cycle are also stored in system variables. A cycle-specific distinction does not therefore have to be made between the addresses. The shell cycle assigns the meaning to the parameters (e.g. for G76, the address F means pitch and not feed). For G70, the feed, speed and tool selection commands (F, S, T) from the program section of the contour definition are relevant.

The same shell cycle is always called in both cases.

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Stock removal cycle, transverse axis G72

Parameters allowed: see G71 (stock removal cycle, longitudinal axis)

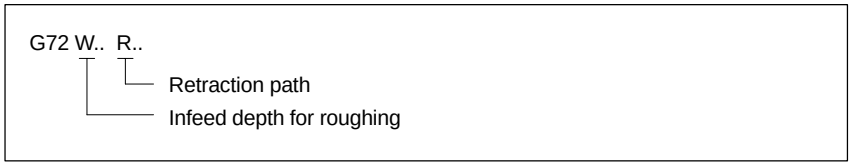


Fig. 2-9 Description of parameters allowed; saving values in GUD

Contour repetition G73

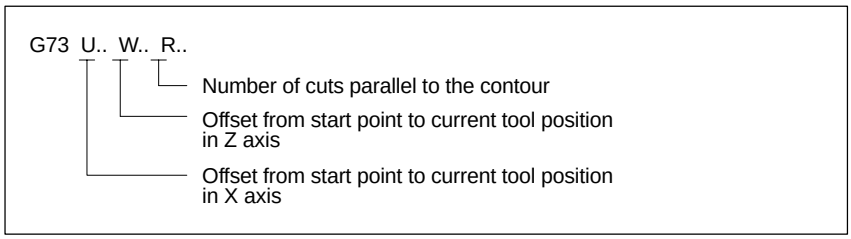


Fig. 2-10 Description of parameters allowed; saving values in GUD

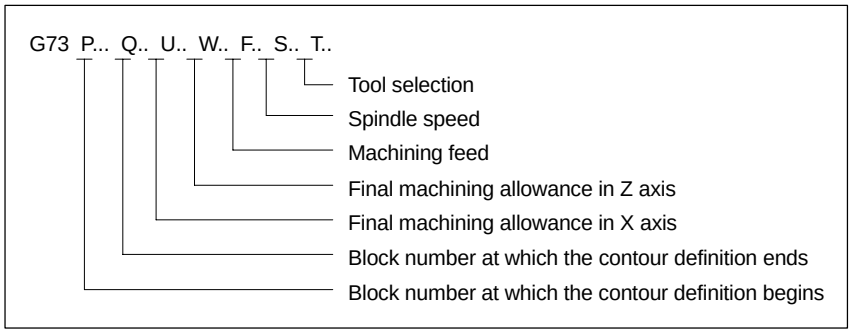


Fig. 2-11 Description of parameters allowed; running cycles

Deep hole drilling and recessing in longitudinal axis G74

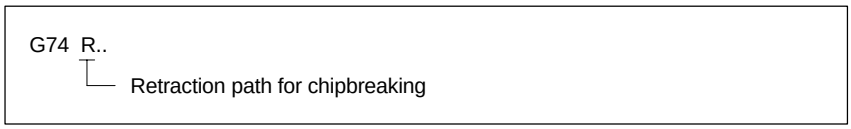


Fig. 2-12 Description of parameters allowed; saving values in GUD

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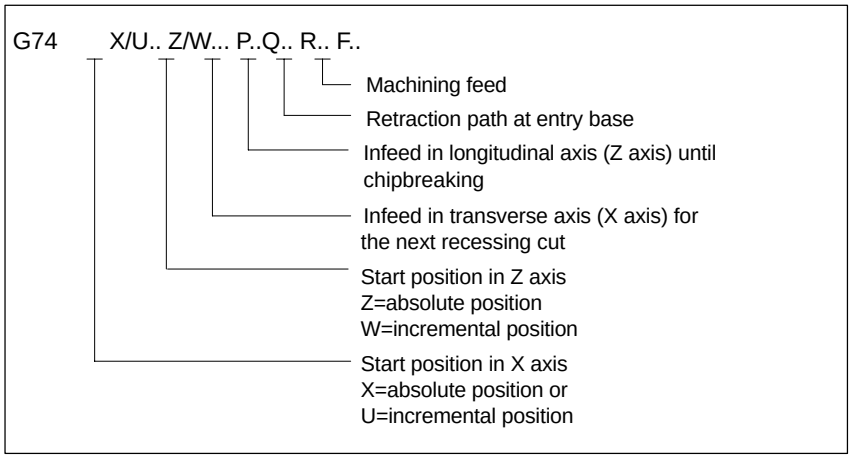


Fig. 2-13 Description of parameters allowed; running cycles

Note

The cycle can be used as a drilling or recessing cycle. If the cycle is used for drilling, addresses X/U and P must not be used.

Deep hole drilling and recessing in transverse axis G75

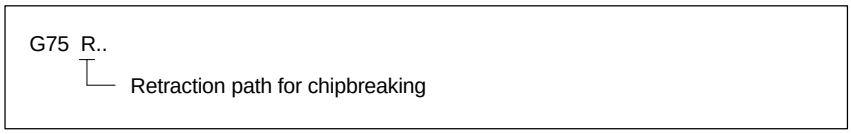


Fig. 2-14 Description of parameters allowed; saving values in GUD

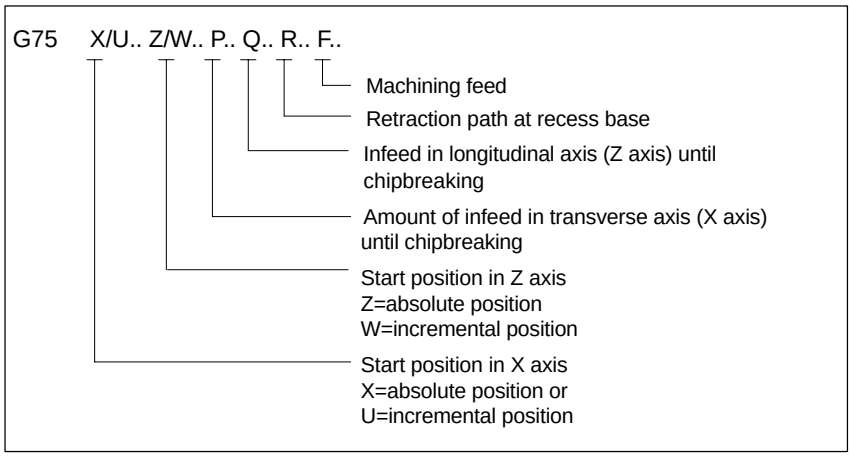


Fig. 2-15 Description of parameters allowed; running cycles

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Note

The cycle can be used as a drilling or recessing cycle. If the cycle is used for drilling, addresses Z/W and Q must not be used.

Multiple thread cutting cycle G76

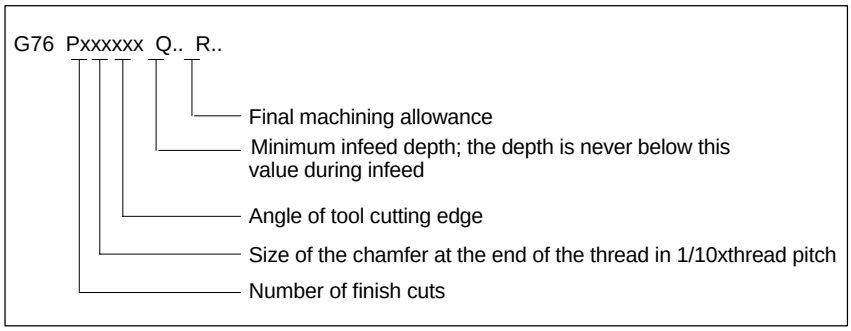


Fig. 2-16 Description of parameters allowed; saving values in GUD

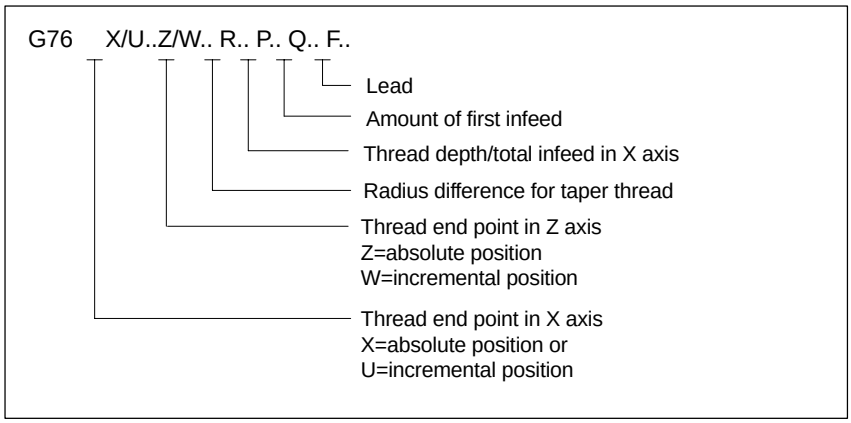
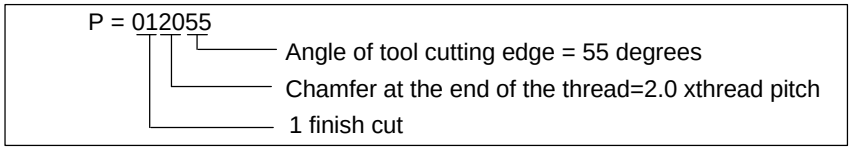


Fig. 2-17 Description of parameters allowed; running cycles

Example for address P:

G76 P012055 Q4 R0.5



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2.4.2 Turning cycles G77 to G79

Table 2-6 Overview of turning cycles G77 to G79

G command	Description
G77	Longitudinal stock removal
G78	Thread cutting
G79	Transverse stock removal

These cycle calls are modal and are called in every NC block containing axis movements. The machining movements are defined in the call parameters after the G function. The following parameters are allowed in NC blocks with cycle calls via G77–G79:

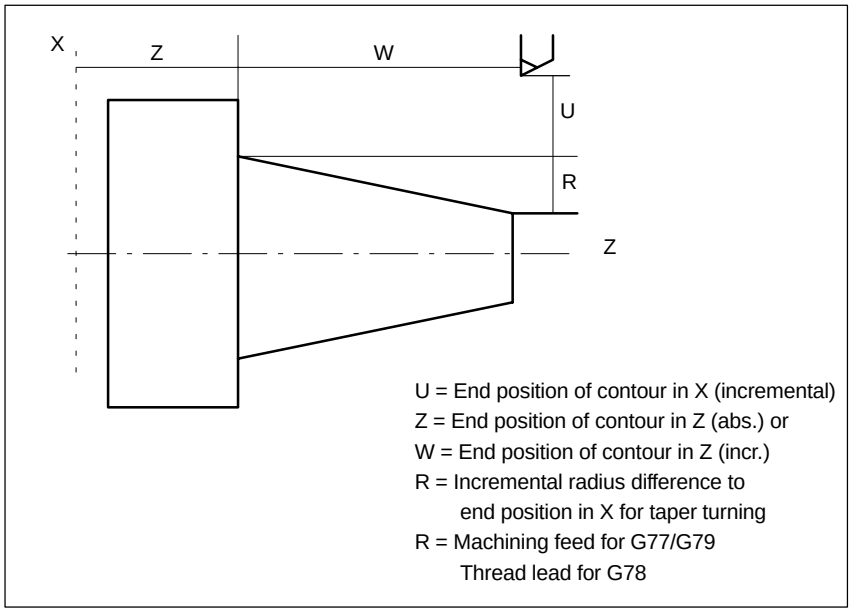


Fig. 2-18 Parameters for cycle calls via G77 to G79

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Longitudinal stock removal G77

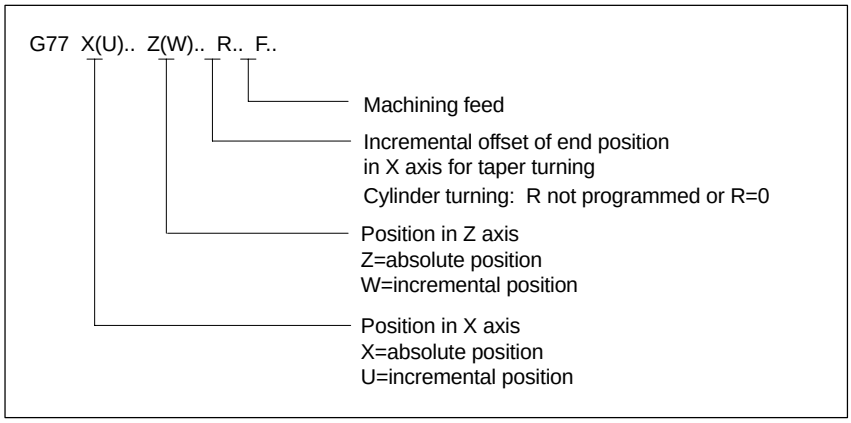


Fig. 2-19 Description of parameters allowed; running cycles

Thread cutting G78

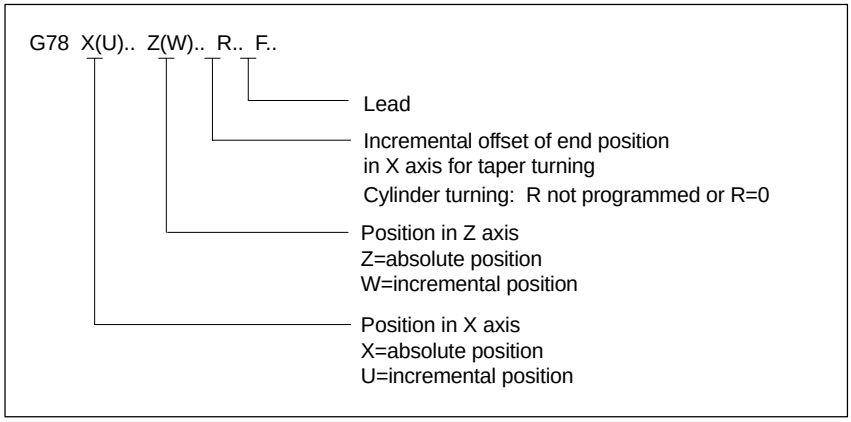


Fig. 2-20 Description of parameters allowed; running cycles

Transverse stock removal G79

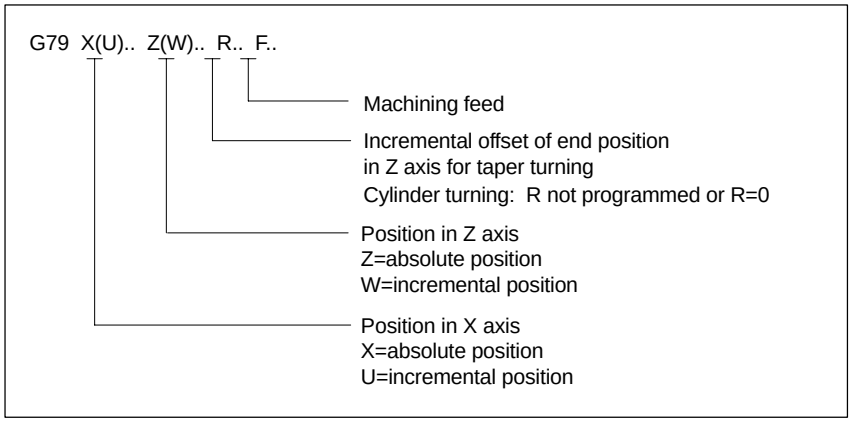


Fig. 2-21 Description of parameters allowed; running cycles

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2.4.3 Drilling cycles G80 to G89

Table 2-7 Overview of drilling cycles

G command	Description
G80	Drilling cycle off
G83	Face deep hole drilling
G84	Face tapping
G85	Face drilling
G87	Side deep hole drilling
G88	Side tapping
G89	Side drilling

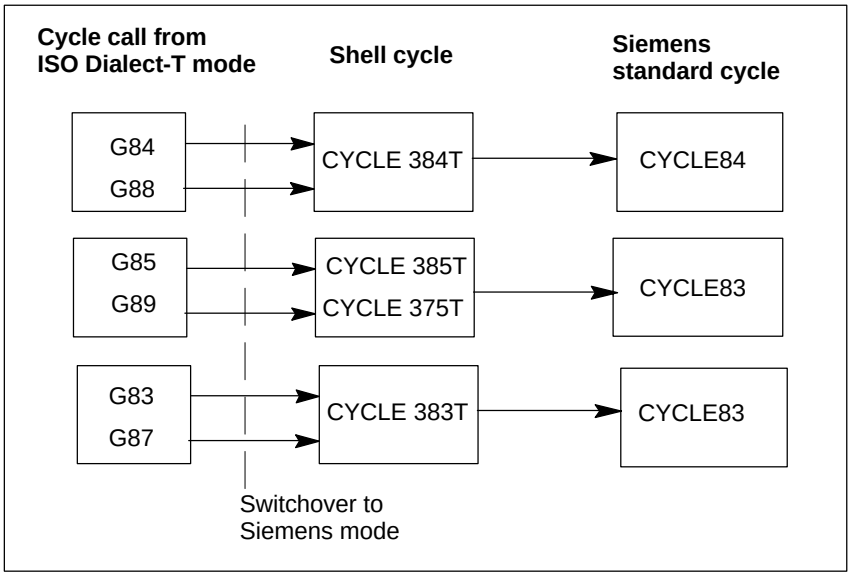


Fig. 2-22 Assignment of the drilling cycle in ISO Dialect-T mode via shell cycle for Siemens standard cycle

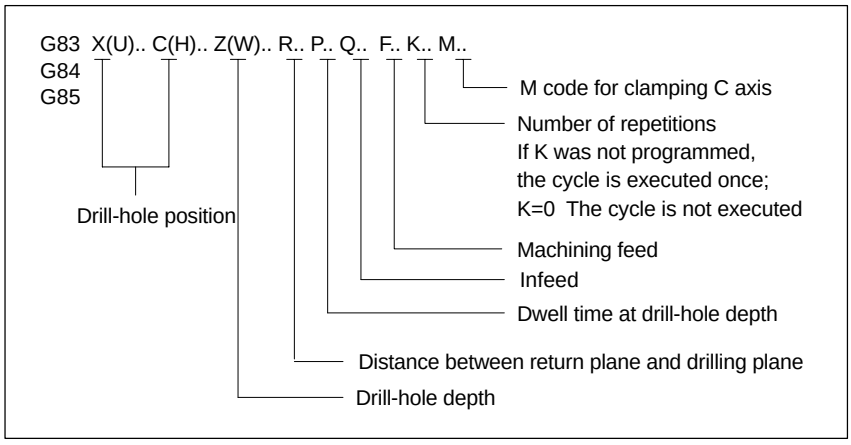


Fig. 2-23 Description of parameters allowed; running cycles

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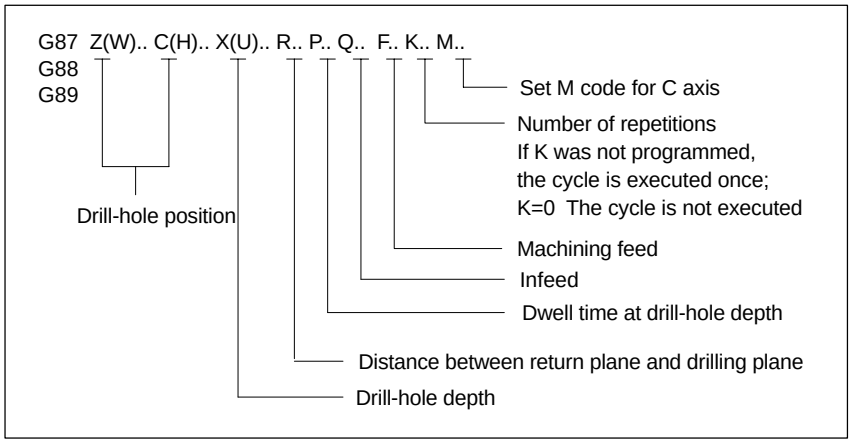


Fig. 2-24 Description of parameters allowed; running cycles

The drilling cycles are modal and are executed in every NC block in which axis movements for axes X, Y, and Z are programmed. While a drilling mode is active, you only have to program the new parameters in order to make parameter modifications. The parameters are stored in system variables $C_xx (xx = Nc address) which are read by the cycles.

The cycle is not executed if a G function of the first G group appears after the cycle G function in the same NC block. Only the axes programmed in the NC block are moved. Addresses R, Q, P, K are not copied into the system variables. The feed programmed in this block is activated.

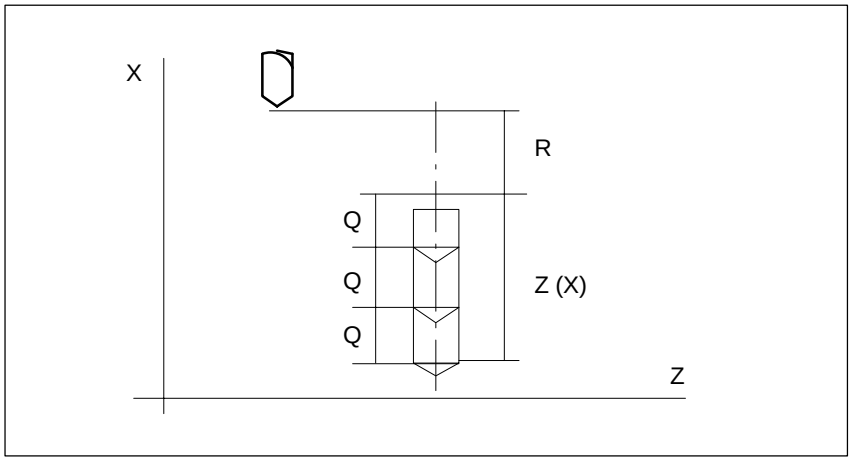


Fig. 2-25 Drilling cycle

Modal cycles

All modal cycles are deselected in ISO Dialect mode with G80 or with a G function of the 1st G group (G00–G03, G33, G34, except for G77–G79).

Cycle parameters can be programmed in the following blocks while a modal cycle is active. These parameters are copied into the system variables so that the shell cycle uses the modified parameters.

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Example:

```
N10 G81 X10. Z15. R5 Q4 P10 F1000
N20 X50.                ;Drilling cycle at position X50
```

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2.4.4 Description of shell cycle CYCLE383T

In ISO Dialect-T mode, the call is performed with G commands G83 and G87.

Notes

The direction from initial plane to reference plane must be identical to the direction from reference plane to final depth. Otherwise this results in an error message from the Siemens drilling cycle.

In ISO Dialect-T, the programmer must take into account the safety clearance when defining the reference plane.In Siemens mode, the safety clearance to the R plane can be specified independently.
This possibility has also been implemented for the ISO cycles.GUD _ZSFR[20] can be used if necessary to enter a safety clearance. If the safety clearance was already allowed for when programming the R plane, the value ”NULL” must be entered in GUD_ZSFR[20].

GUD_ZSFR[23] is used to specify whether the dwell time for G95 (only in the case of deep hole drilling) must be executed in seconds or revolutions.

A shell cycle is only permitted to be called from the external G code (G83/G87). A call in Siemens mode (after switching over with G290 and calling CYCLE383T) is not permitted.

If the G83/G87 block contains axis names other than X/Z (U/W), this results in the alarm (61811) “ISO axis name is not permitted”.

GUD_ZSFR[2] is used to determine whether the cycle is executed with chip breaking or chip removal.

In ISO Dialekt, the constant individual depth is maintained until the remainder is removed on the floor of the drilled hole with an infeed. In Siemens mode, the remainder (less 2*individual depth) on the floor of the drilled hole is subdivided into two equal infeeds.

Note

Alarms are listed with their alarm number and description in Chapter “Alarms”.

2.4.5 Description of shell cycle CYCLE384T

In ISO Dialect-T mode, the call is performed with G commands G84 and G88.

Notes

The direction from initial plane to reference plane must be identical to the direction from reference plane to final depth. Otherwise this results in an error message from the Siemens drilling cycle. ISO Dialect does not monitor this.

In ISO Dialect-T, the programmer must take into account the safety clearance when defining the reference plane. In Siemens mode, the safety clearance to the R plane can be specified independently.
This possibility has also been implemented for the ISO cycles. GUD _ZSFR[20] can be used if necessary to enter a safety clearance. If the safety clearance was already allowed for when programming the R plane, the value ”NULL” must be entered in GUD_ZSFR[20].

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A shell cycle is only permitted to be called from the external G code (G84/G88). A call in Siemens mode (after switching over with G290 and calling CYCLE384T) is not permitted.

If the G84/G88 block contains axis names other than X/Z (U/W), this results in the alarm (61811) “ISO axis name is not permitted”.

The drilling speed during retraction can be controlled via GUD _ZSFI[22] (value in %).
Example: _ZSFI[22]=95, the retraction takes place with 95% of the drilling depth.

Note

Alarms are listed with their alarm number and description in Chapter “Alarms”.

2.4.6 Description of shell cycle CYCLE385T

In ISO Dialect-T mode, the call is performed with G commands G85 and G89.

Notes

The direction from initial plane to reference plane must be identical to the direction from reference plane to final depth. Otherwise this results in an error message from the Siemens drilling cycle. ISO Dialect does not monitor this.

In ISO Dialect-T, the programmer must take into account the safety clearance when defining the reference plane. In Siemens mode, the safety clearance to the R plane can be specified independently.
This possibility has also been implemented for the ISO cycles. GUD _ZSFR[20] can be used if necessary to enter a safety clearance. If the safety clearance was already allowed for when programming the R plane, the value ”NULL” must be entered in GUD_ZSFR[20].

A shell cycle is only permitted to be called from the external G code (G85/G89). A call in Siemens mode (after switching over with G290 and calling CYCLE385T) is not permitted.

If the G85/G89 block contains axis names other than X/Z (U/W), this results in the alarm (61811) “ISO axis name is not permitted”.

If the G84 block contains axis names other than X/Z (U/W), this results in the alarm (61811) “ISO axis name is not permitted”.

Note

Alarms are listed with their alarm number and description in Chapter “Alarms”.

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2.5 System variables

The names of the system variables all begin with $C_ xx. The NC address, whose value is stored in the system variable, appears in the name extension xx. The G number used to call a cycle is always stored in variable $C_G.

For all addresses, bit 0 is set in system variables $C_x_PROG if the address is programmed.

In addition, bit 1 is set in variable $C_x_PROG for axis addresses programmed incrementally.

$C_x_PROG is set to FALSE at the end of the subprogram (M17, RET).

Example 1:

```
N10 G01 G81 X100. Z–50. R20 F100
```

Shell cycle CYCLE381M for G81 is called automatically. The calculations are performed in the shell cycle and the Siemens standard drilling cycle CYCLE82 is then called. The G01 command is not required.

The values of the programmed addresses are written into the following system variables:

Address X is written to system variable $C_X;
Address Z is written to system variable $C_Z;
Address R is written to system variable $C_R;
Address F is written to system variable $C_F;

Example 2:

Axis Z is programmed in incremental dimensions (G91) → $C_Z_PROG=3
Axis Z is programmed in absolute dimensions (G90) → $C_Z_PROG =1

Example 3: Siemens shell cycle for Gxy

```
N10 PROC CYCLE377 DISPLOF     ;Block display remains at G77 block,
                              ;Freeze G code display
N20 DEF REAL DELTA_X, pos_X, pos_Z, FEED
N30 DEF BOOL R_prog, X_prog, Z_prog
N50 DELTA_X = 0
N60 IF $C_R_PROG              ;Only load DELTA_X if address R
N70  DELTA_X = $C_R           ;was programmed
N75 ENDIF
N110 CYCLE...(DELTA_X, $C_X, $C_Z, $C_R_PROG, $C_X_PROG,
$C_Z_PROG, $C_F)              ;Call Siemens cycle
N230 RET                      ;End of shell cycle
```

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Table 2-8 List and description of system variables

Identifier	Type	Description
$C_A	REAL	Value of programmed address A in ISO-Dialect mode for cycle programming
$C_B	REAL	Value of programmed address A in ISO-Dialect mode for cycle programming
....	....	.....
$C_G	INT	G number for cycle calls in external mode
$C_H	REAL	Value of programmed address H in ISO-Dialect mode for cycle programming
$C_I[...]	REAL	Value of programmed address I in ISO-Dialect mode for cycle programming and macro programming with G65/G66. Up to 10 items are possible in one block for macro programming. The values are stored in the array in the order in which they are programmed.
$C_J[...]	REAL	For description see $C_I[...]
$C_K[...]	REAL	For description see $C_I[...]
$C_L	REAL	Value of programmed address L in ISO-Dialect mode for cycle programming
....	....	....
$C_Z	REAL	Value of programmed address Z in ISO-Dialect mode for cycle programming
$C_TS	STRING	String of tool name programmed at address T
$C_A_PROG	INT	Address A is programmed in a block with a cycle call. 0 = not programmed 1 = programmed (absolute) 3 = programmed (incremental)
$C_B_PROG	INT	Address B is programmed in a block with a cycle call. 0 = not programmed 1 = programmed (absolute) 3 = programmed (incremental)
....	....	....
$C_G_PROG	INT	The shell cycle call is programmed with a G function
$C_Z_PROG	INT	Address Z is programmed in a block with a cycle call. 0 = not programmed 1 = programmed (absolute) 3 = programmed (incremental)
$C_TS_PROG	INT	A tool name was programmed at address T TRUE=programmed, FALSE=not programmed
$C_ALL_PROG	INT	Bitmap of all programmed addresses in a block with a cycle call Bit 0 = address A Bit 25 = address Z Bit = 1 address programmed in incremental dimensions Bit = 0 address not programmed
$P_EXTGG[n]	INT	Active G code of the external language
$C_INC_PROG	INT	Bitmap of all programmed incremental addresses in a block with a cycle call Bit 0 = address A Bit 25 = address Z Bit = 1 address programmed in incremental dimensions Bit = 0 address programmed in absolute dimensions
$C_I_NUM	INT	Cycle programming: Value is always 1 if bit 0 set in $C_I_PROG. Macro programming: Number of I addresses programmed in block (max. 10).
$C_J_NUM	INT	For description see $C_I_NUM
$C_K_NUM	INT	For description see $C_I_NUM
$P_AP	INT	Polar coordinates 0 = OFF 1 = ON

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2.6 Programming contour definitions (ISO Dialect-T)

Contour definitions can be programmed in both ISO Dialect-T mode and Siemens mode.

There are 3 basic shapes of contour

- A straight line
  The end point is programmed with a Cartesian coordinate and an angle
- Two straight lines
  The transition is programmed with a rounding or chamfer
- Three straight lines
  The transitions are programmed with a rounding or chamfer

In the descriptions below, indices are occasionally assigned to the address letters X, Z, A, R and C to establish a unique assignment between the NC block and the associated drawing. These indices do not appear in the NC program. The assignment is always derived uniquely from the block containing the address letter. Address letter Q is used as a placeholder for R or C, where either of these letters can appear. Q can also be omitted. In this case, no chamfer or rounding is inserted at the transition between the two linear sections.

Any number of other NC addresses can be used in blocks defining contours, e.g. address letters for further axes (individual axes or an axis perpendicular to the machining plane), auxiliary function parameters, G codes, velocities, etc.

In the following examples, it is assumed that G18 is active. Programming of contours is also possible without restriction with G17 or G19, however.

ISO Dialect mode

Address C is used in ISO Dialect mode both as an axis identifier and as an identifier for a chamfer on the contour.

Address R can be a cycle parameter or an identifier for the radius in a contour.

In order to distinguish between these two options, a “,” must be placed in front of the C or R address during contour definition programming (as in ISO Dialect). A comma does not have to be entered if an angle is programmed before C or R. If a radius and a chamfer are programmed together in the same block, e.g. N333 X100 A10 C20 R15, regardless of the programming sequence, a radius will always be inserted in the contour. The chamfer is ignored.

Siemens mode

The identifiers for angle, radius and chamfer are defined by machine data in Siemens mode. This prevents the occurrence of name conflicts. A comma must not be programmed before the identifier for radius or chamfer.

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Note

MD 10652 for angle:	$MN_CONTEUR_DEF_ANGLE_NAME
MD 10654 for radius:	$MN_RADIUS_NAME
MD 10656 for chamfer:	$MN_CHAMFER_NAME

2.6.1 End point programming with angles

If address letter A appears in an NC block, one, both or none of the axes on the active plane can be programmed.

If no axis on the active plane is programmed, the block is either the first or the second block of a contour totalling two blocks. If the block is the second block of such a contour definition, the start and end point in the active plane are identical. In this case, the contour merely consists of a movement perpendicular to the active plane.

If one axis on the active plane is programmed, either a single straight line is being described, whose end point is determined exactly by the angle and the programmed Cartesian coordinate, or we are dealing with the second block of a contour totalling two blocks. In the latter case, the missing coordinate will be set equal to the last (modal) position reached.

If two axes on the active plane are programmed, the block is the second block of a contour comprising two blocks. If the current block was not preceded by a block with angle programming without programmed axes of the active plane, the block is not permitted.

Angle A must only be programmed with linear or spindle interpolation. (Spline interpolation only in Siemens mode.)

Alarms are generated in the following situations:

- In a contour consisting of two blocks, the active plane was changed during the transition from the first to the second block.
- In a contour consisting of two straight lines, a valid intermediate point cannot be generated from the programmed angles.
- Neither linear nor spline interpolation is active in a block with address A.
- A block with address A without a programmed axis on the active plane is not followed by a block with which the end point of the contour can be determined. This is the case if the block is the last block in a program or if the following block contains a preprocessor stop.
- No angle was programmed in the second block of a contour consisting of two straight lines.
- Both axes on the active plane are programmed in a block with address A which is not the second block of a contour consisting of two straight lines.
- Programmed Cartesian coordinate and programmed angle are incompatible.

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2.6.2 Straight line with angle

The end point is defined by specifying the angle A and one of the two coordinates X_2 or Z_2.

Programming syntax:

X_2.. A.. or
Z_2.. A..

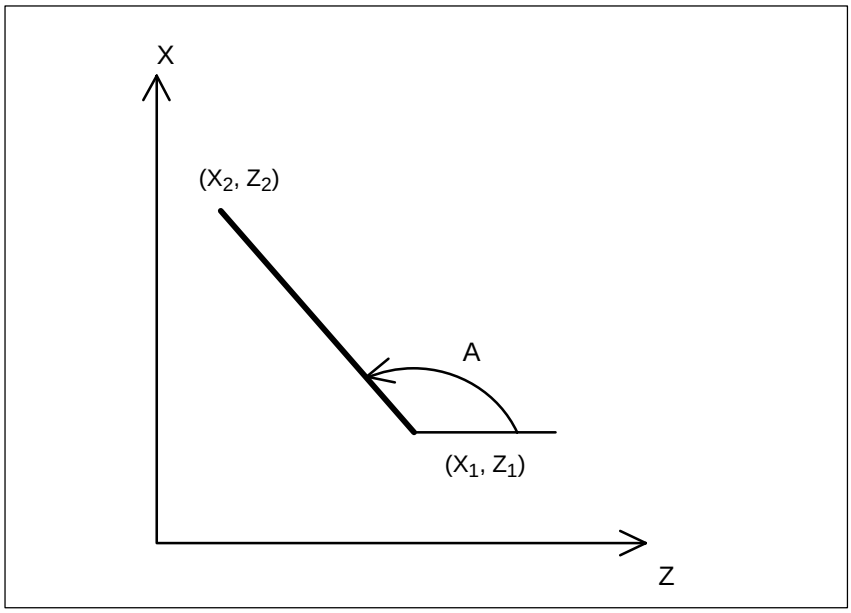


Fig. 2-26 Straight line with angle

Example (Fig. 3-24):

Programming in ISO Dialect-T mode:

```
N10 G1 X5. Z70. F1000 G18
N 20 X88.8 A 110 or (Z39.5 A110)
```

Programming in Siemens mode:

```
N10 X5. Z70. F1000 G18
N20 X88.8 ANG=110 or (Z39.5 ANG=110)
```

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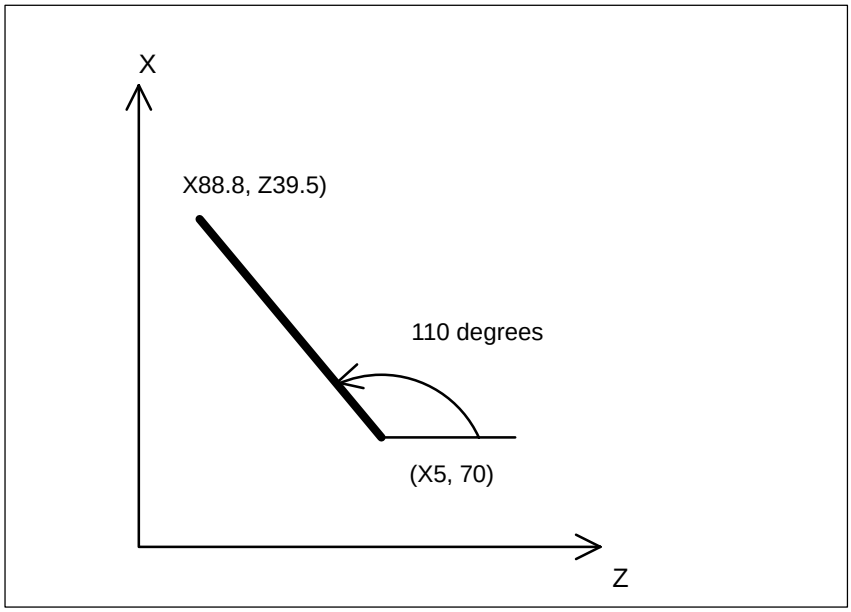


Fig. 2-27 Straight line with angle

2.6.3 Two straight lines

The end point of the first straight line can be programmed either by specifying the Cartesian coordinates or by specifying the angle of the two straight lines relative to the abscissa.

Programming syntax:

```
N10  A1..   (Q..)
N20  X3..  Z3..  A2..
```

or

```
N10  X1.. Z1..  (Q..)
N20  X3..  Z3..
```


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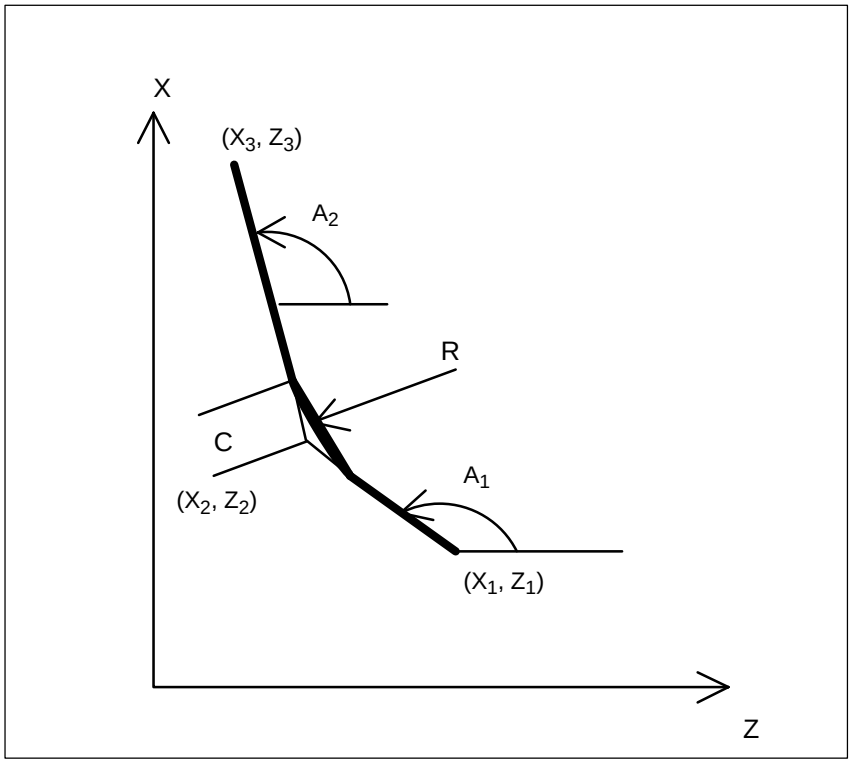


Fig. 2-28 Two straight lines

Example (Fig. 3-26):

Programming in ISO Dialect-T mode:

```
N10 G1 X10. Z80. F1000 G18
N20 A 1.48.64 C5.5
N30 X85. Z40. A100
```

Programming in Siemens mode:

```
N10 X10. Z80. F1000 G18
N20 ANG=148.65 CHR=5.5
N30 X85. Z40. ANG=100
```

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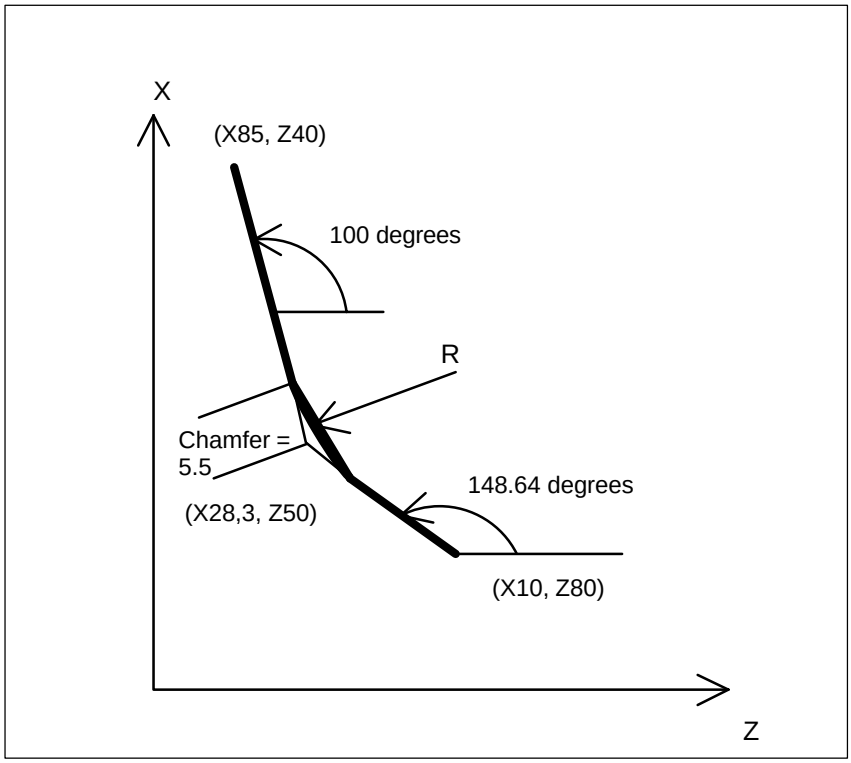


Fig. 2-29 Two straight lines

2.6.4 Three straight lines

The end point of the third straight line must be programmed with Cartesian coordinates. Either a chamfer or a radius can be used for the transition from the second to the third coordinate.

This type of programming can be used for any number of further blocks, i.e. a distinction does not have to be made between contours with two or more blocks.

Programming syntax:

```
N10  X2..  Z2..   (Q1..)
N20  X3..  Z3..   (Q2..)
N30  X4..  Z4..
```

or

```
N10  A1..  (Q1..)
N20  X3..  Z3..  A2..  (Q2..)
N30  X4..  Z4..
```

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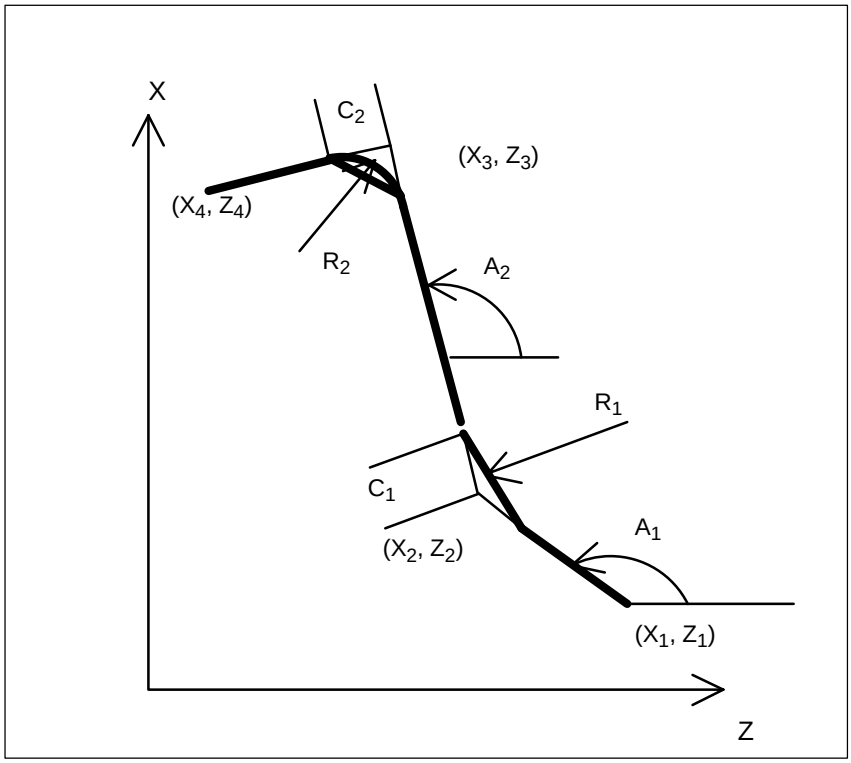


Fig. 2-30 Three straight lines

Example (Fig. 3-29):

Programming in ISO Dialect-T mode:

```
N10 G1 X10. Z100. F1000 G18
N20 A140 C7.5
N30 X80. Z70. A95.824, R10
```

Programming in Siemens mode:

```
N10 X10. Z100. F1000 G18
N20 ANG=140 CHR=7,5
N30 X80. Z70. ANG=95.824 RND=10
```

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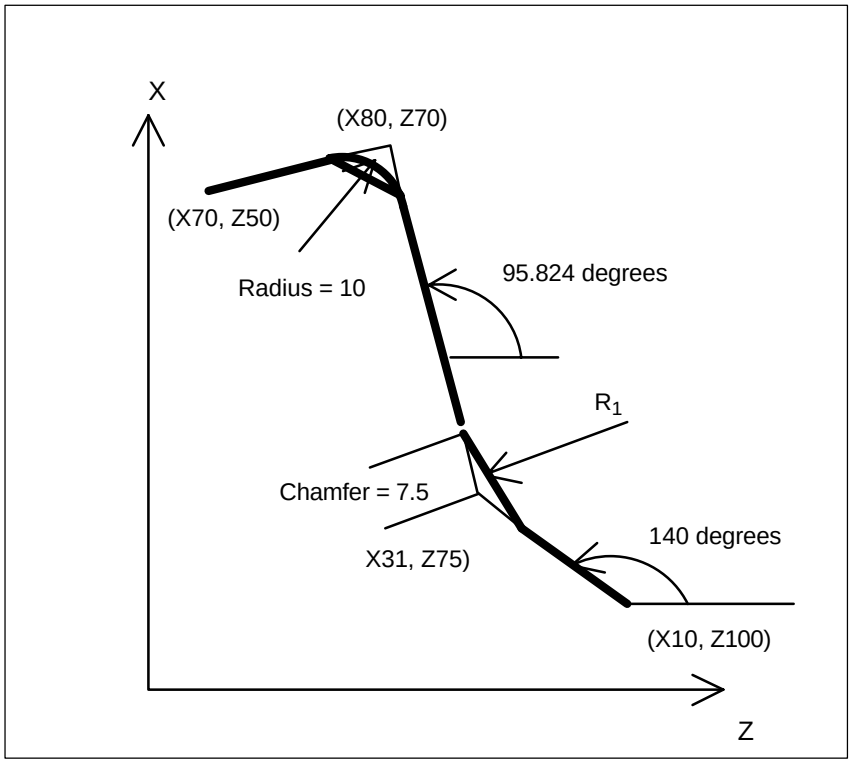


Fig. 2-31 Three straight lines

■

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Boundary Conditions

3

Availability of the "ISO Dialect" function

The function is an option and available for

- SINUMERIK 810D with CCU1 and CCU2
- SINUMERIK 840D with NCU572.2 and NCU573.2

3.1 Restrictions

The following section describes functions for which the SINUMERIK 840D exhibits **non-compatibility** with the ISO Dialect-Original when in ISO Dialect mode.

Mode switchover

The standard set of machine data implements "Siemens" mode only. No external NC language is generated as a 2nd G code table.

Machine data $MN_NC_USER_CODE_CONF_NAME_TAB is only valid for NC language commands in Siemens mode.

For compatibility with Siemens mode (with reference to machine data input, OPI interface: "data array"[0] = 1st G group), the zero G groups defined in ISO Dialect-Original are converted as follows in ISO Dialect mode:

ISO–Dialect–M: G Group 0 → G Group 18
ISO–Dialect–T: G Group 0 → G Group 17

Implicit mode change

Asynchronous subprograms (ASUPs), INI files and macro/GUD definition files always run in Siemens mode. If necessary, an implicit change to Siemens mode is executed. When the file has finished running, the original mode of the external CNC system is restored.

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3.1.1 Program commands

F value

ISO Dialect-M has fixed F values which can be selected with F1 to F9. These fixed values are not available in ISO Dialect0 mode. F1 to F9 are interpreted as value F1 to F9.

G02/G03

Programming G02/G03 without a radius parameter results in G01 in ISO Dialect-T-Original, and in a full circle with an undefined radius in ISO Dialect-M-Original. In both cases, an NC alarm is output in ISO Dialect mode.

G04 X..

With ISO Dialect-Original, the dwell time in the X axis is displayed as the distance to go. This dwell time is not displayed in ISO Dialect mode; the message “dwell time running” is displayed.

G16

Polar coordinate programming is terminated with G15.
In ISO Dialect-Original, the polar radius and polar angle are retained. The next time G16 is programmed, an incremental movement can be superimposed on the angle. The result is a traversing movement which cannot be evaluated. The angle and the radius are deleted on a reset or M30.

In ISO Dialect mode, the polar radius and polar angle are deleted with G15. With G16, the traversing movement is always performed with an angle and a radius of 0.

Example: Axis U is parallel to axis X
G17 U Y ; Plane U Y is selected instead of X Y.

In ISO Dialect mode, a parallel axis **cannot** be programmed with G17/G18/G19.

G20/G21

In ISO Dialect-Original mode, the zero offsets are not converted on switchover. The decimal point is merely shifted by one decimal place. In ISO Dialect mode, the zero offset values are converted completely.

G22

Protection zone 4 is activated with G22 in ISO Dialect-M-Original. This is not available in phase 1. Protection zones 1/2 and 3 are implemented. Protection zone 4 is entered permanently in the machine data settings for ISO Dialect. This is not possible with the 840 D. In ISO Dialect mode there is only one protection zone.

G40

In ISO-Dialect-T mode, a vector can be programmed in a linear block with I, J, K; the vector controls the behavior at the end of the block. This function is not possible in ISO Dialect-Original mode. An alarm is output if I, J and K are programmed with G40.

G41/G42

The cutter radius compensation functions are not compatible between ISO Dialect-Original and 840D.

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G53

If G53 (position in machine coordinate system approached) is called while G41/G42 is active, the axes are not traversed separately, but interpolated instead.

G63

With ISO Dialect, G63 can appear in every block. The override is disabled in that block. Override 0 also triggers a stop at the start of a G53 block. This function is practical in combination with rigid tapping in G01 interpolation. In ISO Dialect mode, G63 is only effective in a G01 block. If G63 is selected in a G00 block, this has no effect on the block.

G94/G95

Each time you switch from revolutional feedrate (G95) to linear feedrate (G94) or vice-versa, feed F must be programmed again. If the feed is missing, alarm 10860 “No feedrate programmed” is output. ISO Dialect-Original mode interprets the feedrate which has been programmed once as either a revolutional or a linear feedrate when the feedtype is changed.

G96

With ISO Dialect-M the axis to which G96 refers can be programmed with P. This generates an alarm in ISO mode. Only P0 transverse axis is possible from machine data.

M06

Tool-changing cycle: If an M function is assigned to a cycle via MD $MN_M_NO_FCT_CYCLE and $MN_M_NO_FCT_CYCLE_NAME, not all of the parameters programmed in the block can be accessed in the cycle. Only the programmed T number can be read with $C_T/$C_T_PROG. Traversing movements in the M block are executed before the call.

Syntax-governing G functions

With ISO Dialect-Original, several syntax-governing G functions can be programmed in the same block. A distinction is made between the following:

syntax governing → non–modal

syntax governing → modal

non syntax governing → modal

non syntax governing → non–modal

In Siemens mode, more than one syntax-governing G function in a block causes an NC alarm.

Skip

Only skip level 1–8 can be selected. In ISO Dialect-Original, level 9 can also be selected. The skip character “/” must appear at the start of the block. If it appears within the block, this generates an alarm.

3.1.2 Tool management

Tool management, tool life and workpiece count monitoring can be reproduced with the Siemens tool management system.

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Tool data

Milling: Only Tool Compensation Memory C is supported, i.e. multi-column structure of the tool compensation memory (D == H applies).
The variant whereby the tool and offset number are generated from the same value for milling is not supported.

Read-out/archiving of the current tool data in ISO Dialect mode (G10) is not possible. It is not possible to modify the tool data via G10 until the tool offsets have been set up by an operator actio.

Tool length compensation

If tool length compensation input is active in diameter mode, the input for geometry and wear can be configured in diameters with ISO Dialect-Original. In ISO Dialect mode, only the wear is specified as a diameter. The geometry is always specified in radius mode.

With ISO Dialect-Original, the tool length is calculated as a diameter or radius in the transverse axis, depending on a machine parameter.
In Siemens mode, the length offset is always calculated as a diameter, it is not possible to switch over.

Compensation of the tool length offset by shifting the coordinate system is not possible.

The geometry and wear cannot be taken from different compensation memories for the “Turning” technology.

Modified tool offsets are valid when the next T, H or D value is programmed.

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3.1.3 Control system response to Power ON, Reset and block search

Power ON

On Power ON, the Siemens G code list is created for all NC channels with the possible code changes from MD $MN_NC_USER_CODE_CONF_NAME_TAB.

Startup, Reset

MD $MC_GCODE_RESET_VALUES[46] defines the startup and reset behavior depending on MD $MN_EXTERN_CNC_SYSTEM.

This decides between the G codes of ISO Dialect-M and ISO Dialect-T.

The change between Siemens and ISO Dialect mode has no effect on the modal G functions while the program is running.

Block search

The “search to end of block” block search type available in Siemens mode is the equivalent of the search behavior with ISO Dialect.

In “block search without calculation” mode, the user must provide an appropriate search target, especially in NC programs with a mode switchover (e.g. an NC block with a command from G group 47).

■

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Notes

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